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Department of Electrical & Electronics
School of Engineering, The Federal Polytechnic Mubi PMB 35
Adamawa State

Course Lecturers:
Dr. A.liyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE
Mr. Mohammed Bobboi

LECTURE NOTES 1: **WAVE EQUATION**
WAVEGUIDES
WAVEGUIDES COMPONENTS
MICROWAVE TUBES
MICROWAVE SEMICONDUCTORS
OPTICAL FIBRE

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INTRODUCTION ELECTROMAGNETIC WAVE

Wave Equation Recap

A plane wave only has dependence (and, therefore, propagation) in one direction. In the figure below, the plane wave is propagating in the z direction. Notice that the polarization is in the y direction (thus, it is $Ey(z)$), but the wave is propagating only in the z direction.

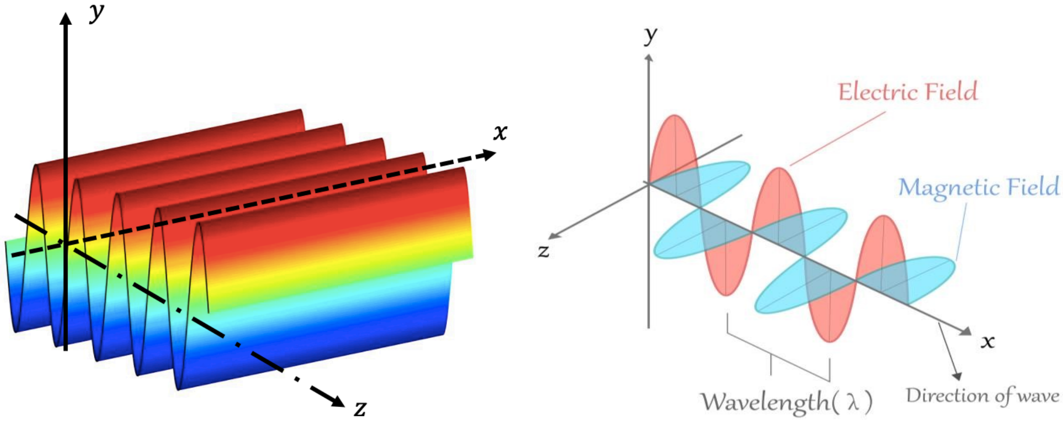


Figure 1. A plane wave E$_y$(z) has no x or y dependence.

We define the **wavelength** λ to be the distance between successive peaks. Since the wave has no dependence in the x or y direction, we can simplify the description of the wave as follows

$$E_y(x, y, z, t) = E_y(z, t) \quad \text{(Equ. 1)}$$

If the wave time harmonic is known, the z and t dependence can be separated as follows:

$$E_y(z, t) = E_y(z)e^{j\omega t} \quad \text{(Equ. 2)}$$

Let's investigate whether (and how) this function will satisfy the wave equation. First, we'll determine the second derivative of E_y with respect to z:

$$\frac{\partial^2 E_y(z,t)}{\partial z^2} = \frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t} \quad \text{(Equ. 3)}$$

Next, we'll calculate the second derivative with respect to t:

$$\frac{\partial^2 E_y(z,t)}{\partial t^2} = E_y(z)(j\omega)^2 e^{j\omega t} \quad \text{(Equ. 4)}$$

Substituting Equations 3 and 4 into the wave equation 2 above, we find:

$$\frac{\partial^2 E_y(z)}{\partial z^2}e^{j\omega t} - \frac{1}{c^2}E_y(z)(j\omega)^2 e^{j\omega t} = 0$$ (Equ. 5)

This can be simplified as follows:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + \left(\frac{\omega}{c}\right)^2 E_y(z) = 0$$ (Equ. 6)

If we define k (called the **wave number**) as follows:

$$k = \frac{\omega}{c}$$ (Equ. 7)

Then Equation 6 simplifies to:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + k^2 E_y(z) = 0$$ (Equ. 8)

This is known as the **one-dimensional Helmholtz equation**, and it is a well-known differential equation with a well-known solution:

$$E_y(z) = ae^{-jkz} + be^{jkz}$$ (Equ. 9)

Adding the time dependence back into this result, we obtain:

$$E_y(z,t) = ae^{-jkz}e^{j\omega t} + be^{jkz}e^{j\omega t}$$ (Equ. 10)

Which can be written as:

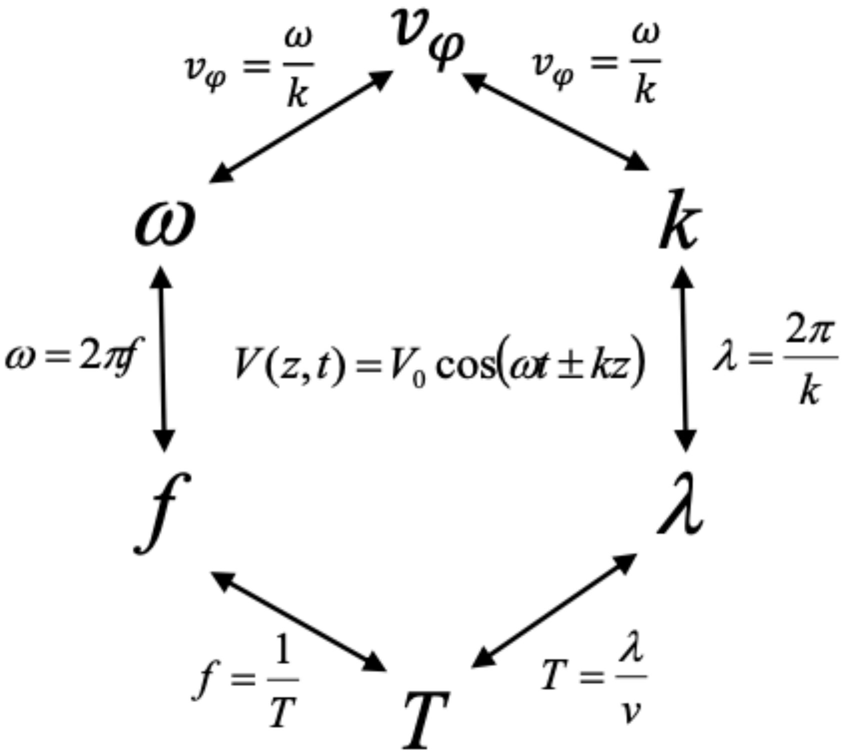
$$E_y(z,t) = ae^{j(\omega t - kz)} + be^{j(\omega t + kz)}$$ (Equ. 11)

Notice that this function has the form of Equation 2 as long as Equation 1 is followed, so it is guaranteed to be a solution to the wave equation. In order to ensure that Equation 11 is a solution to the wave equation, the following relationship must be true.

$$v_\phi = \frac{dz}{dt} = \pm c = \pm\frac{\omega}{k}$$ (Equ. 12)

The positive signs correspond to a right-moving wave, and the negative signs to a left-moving wave. We now have several variables that describe the time and spatial variation of

$E(z,t)$: wavelength λ, wave number k, radial frequency ω, frequency f, period T, and phase velocity $v\phi$. Figure 2 below is designed to graphically demonstrate the relationships among these six variables. If you know two of the variables, you will be able to calculate the other four.



PROPAGATION THROUGH PARALLEL PLANE WAVEGUIDE

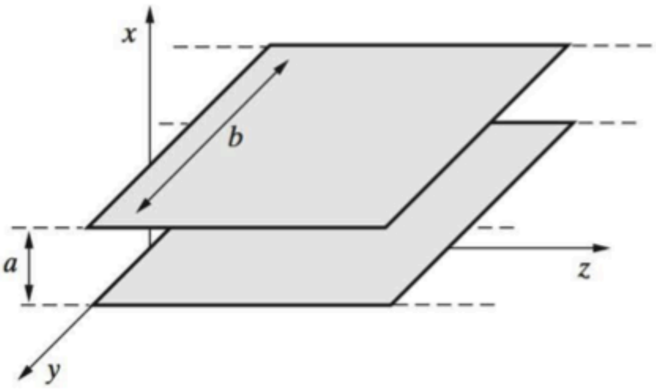


Figure 2 Parallel-plate waveguide.

In an ideal parallel-plate waveguide, the plates are assumed to be infinite in extent in the y and z directions. In practice, the width b of the guide in the y direction is finite but is typically much larger than a. The material between the plates is assumed to be a lossless dielectric.

As shown in Figure 2. The plates have an infinite extent in the z direction, in which the waves are to be guided. In practice the conductor plates would have a finite extent b in the y direction, but it is typically much larger than the plate separation a, so the effects of fringing fields at the edges are negligible. For our purposes here, we assume that the conductor plates have infinite extent $(b >> a)$ in the y direction, so the electric and magnetic fields do not vary with y. Just as in the case of waves propagating in unbounded media, the configuration and propagation of the electromagnetic fields in any guiding structure are governed by Maxwell's equations and the wave equations derived from them. However, in the case of metallic waveguides, the solutions of the wave equations are now subject to the following boundary conditions at the conductor surfaces:

$$E_{tangential} = 0 \qquad\qquad H_{normal} = 0$$

EXPRESSION FOR THE FIELD E & H VECTORS

As presented in the fig. 2 above, assuming that the medium between the perfectly conducting plates is source-free, simple (i.e., ϵ and μ are simple constants), and lossless. (i.e., $a = 0$, μ and ϵ are real), the governing equations are the time-harmonic phasor form of Maxwell's equations, namely

$$\nabla \times \boldsymbol{H} = j\omega\epsilon\boldsymbol{E} \quad \text{(Eqn. 13.1a)}$$

$$\nabla \cdot E = 0 \quad \text{(Eqn. 13.1b)}$$

$$\nabla \times \boldsymbol{E} = -j\omega\epsilon\mu\boldsymbol{H} \quad \text{(Eqn. 13.1c)}$$

$$\nabla . \mathrm{H} = 0 \quad \text{(Eqn. 13.1d)}$$

from which the wave equations for $\boldsymbol{E}$ and $\boldsymbol{H}$ can be derived in a straightforward manner as

$$\nabla^2\boldsymbol{E} - (j\beta)^2\boldsymbol{E} = \boldsymbol{0} \quad \text{(Eqn. 14.a)}$$

$$\nabla^2\boldsymbol{H} - (j\beta)^2\boldsymbol{H} = \boldsymbol{0} \quad \text{(Eqn. 14b)}$$

where $\beta = \omega\sqrt{\epsilon\mu}$

Equations (13.1a) and (13.1c) can be written in component form for rectangular coordinates (note that the conductor configuration is rectangular; otherwise, for example for circularly shaped conductors, it is more appropriate to write the equations in terms of cylindrical coordinates r, $\emptyset$, and *z as detail under waveguide below)* for the region between plates (assumed to be a lossless dielectric, with both α and ϵ'' equal to zero) as:

$$\underbrace{\begin{aligned} \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} &= j\omega\epsilon E_x \\ \frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} &= j\omega\epsilon E_y \\ \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} &= j\omega\epsilon E_z \end{aligned}}_{\nabla\times\mathbf{H}=j\omega\epsilon\mathbf{E}} \qquad \underbrace{\begin{aligned} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= -j\omega\mu H_x \\ \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} &= -j\omega\mu H_y \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= -j\omega\mu H_z \end{aligned}}_{\nabla\times\mathbf{E}=-j\omega\mu\mathbf{H}} \quad \text{(Eqn. 15)}$$

Note that since the extent of the conductor plates in the y direction is infinite, none of the field quantities vary in the y direction, and as a result, the $\partial(\cdots)/\partial y$ terms in (15) are all zero.

We assume propagation in the $+z$ direction, so all field components vary as $e^{-\bar{\gamma}z}$, where $\bar{\gamma} = \bar{\alpha} + j\,\bar{\beta}$, with $\bar{\alpha}$ and $\bar{\beta}$, being real. We consider the two special cases for which $\bar{\beta} = 0$, or for which $\bar{\alpha} = 0$. When the phasor fields vary in this manner, the real fields vary as

$$\mathcal{Re}\{e^{-\bar{\gamma}z}e^{j\omega t}\} = \begin{cases} e^{-\bar{\alpha}z}\cos(\omega t) & \bar{\gamma} = \bar{\alpha} \quad \text{Evanescent wave} \\ \cos(\omega t - \bar{\beta}z) & \bar{\gamma} = j\bar{\beta} \quad \text{Propagating wave} \end{cases}$$

An ***evanescent*** wave is one that doesn't propagate, instead exponentially decays with distance. The wave equation(s) (15) can be rewritten as

$$\nabla^2\mathbf{E} - (j\beta)^2\mathbf{E} = 0 \quad \rightarrow \quad \frac{\partial^2\mathbf{E}}{\partial x^2} + \frac{\partial^2\mathbf{E}}{\partial y^2} + \frac{\partial^2\mathbf{E}}{\partial z^2} - (j\beta)^2\mathbf{E} = 0$$

$$\nabla^2\mathbf{H} - (j\beta)^2\mathbf{H} = 0 \quad \rightarrow \quad \frac{\partial^2\mathbf{H}}{\partial x^2} + \frac{\partial^2\mathbf{H}}{\partial y^2} + \frac{\partial^2\mathbf{H}}{\partial z^2} - (j\beta)^2\mathbf{H} = 0$$

Once again, we assume that none of the field quantities vary in the y direction. In other words, we have

$$\frac{\partial^2}{\partial y^2}(\cdots) = 0$$

The field variations in the x direction cannot be similarly limited because the perfect conductors impose boundary conditions in the x direction at $x = 0$ and $x = a$. Note

that for any of the field components, for example, for H_y, we can explicitly show the variations in the z direction by expressing $H_y\,(x, z)$ as a product of two functions, one having x and the other z, respectively, as their independent variables:

$$H_y(x, z) = H_y^0(x)e^{-\bar{\gamma}z}$$

where $H_y^0(x)$ is a function only of x. We then have

$$\frac{\partial}{\partial z}H_y(x, z) = -\bar{\gamma}H_y^0(x)e^{-\bar{\gamma}z} = -\bar{\gamma}H_y(x, z) \quad \longrightarrow \quad \frac{\partial}{\partial z} \quad \rightarrow \quad -\bar{\gamma}$$

Thus, equations (Eqn. 14) can be rewritten as

$$\frac{\partial^2 \mathbf{E}}{\partial x^2} + \bar{\gamma}^2 \mathbf{E} - (j\beta)^2 \mathbf{E} = 0$$ (Eqn. 16)

$$\frac{\partial^2 \mathbf{H}}{\partial x^2} + \bar{\gamma}^2 \mathbf{H} - (j\beta)^2 \mathbf{H} = 0$$ (Eqn. 17

Substituting $\partial/\partial z \rightarrow -\bar{\gamma}$ and $\partial/\partial y = 0$ in (15), we find

$$\begin{aligned} \bar{\gamma} E_y &= -j\omega\mu H_x & \bar{\gamma} H_y &= j\omega\epsilon E_x \\ -\bar{\gamma} E_x - \frac{\partial E_z}{\partial x} &= -j\omega\mu H_y & -\bar{\gamma} H_x - \frac{\partial H_z}{\partial x} &= j\omega\epsilon E_y \\ \frac{\partial E_y}{\partial x} &= -j\omega\mu H_z & \frac{\partial H_y}{\partial x} &= j\omega\epsilon E_z \end{aligned}$$ (Eqn. 18)

The foregoing relationships between the field components are valid in time-harmonic electromagnetic wave solution for which:

(i) There are no y direction (i.e., $\partial(\cdot)/\partial y = 0$) and

(ii) The field components vary in the z direction as $e^{-\bar{\gamma} z}$.

In general, if a solution of the wave equation subject to the boundary conditions is obtained for any one of the field components, the other components can be found from Equation 18. In some cases, it is convenient to express the various field components explicitly in terms of the axial components (i.e., E_z and H_z). Introducing

$$h^2 = \bar{\gamma}^2 - (j\beta)^2$$ (Eqn. 19)

we can rewrite equations (18) in the form

$$H_x = -\frac{\bar{\gamma}}{h^2} \frac{\partial H_z}{\partial x}$$ (Eqn. 20 a)

$$H_y = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}$$ (Eqn. 20 b)

$$E_x = -\frac{\bar{\gamma}}{h^2} \frac{\partial E_z}{\partial x}$$ (Eqn. 20 b)

$$E_y = +\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$ (Eqn. 20 d)

These expressions are useful because the different possible solutions of the wave equation are categorized in terms of the axial components (i.e., z components), as discussed in the next section.

FIELD SOLUTIONS FOR TE TM, AND TEM WAVES MODES

In general, all the field components, including both E_z and H_z, may need to be nonzero to satisfy the boundary conditions imposed by an arbitrary conductor or dielectric structure. However, it is convenient and customary to divide the solutions of guided-wave equations into three categories:

$$\left.\begin{aligned} E_z &= 0 \\ H_z &\neq 0 \end{aligned}\right\} = \text{Transverse electric (TE) waves}$$

$$\left.\begin{aligned} E_z &\neq 0 \\ H_z &= 0 \end{aligned}\right\} = \text{Transverse magnetic (TM) waves}$$

$$E_z, H_z = 0 \quad \longrightarrow \quad \text{Transverse electromagnetic (TEM) waves}$$

where z is the axial direction of the guide along which the wave propagation takes place. In the following, we separately examine the field solutions for **TE, TM,** and **TE**M waves.

Transverse Electric (TE) Waves.

We first consider transverse electric (TE) waves, for which we have $E_z = 0$, $H_z \neq 0$. With $E_z = 0$, the electric field vector of a **TE** wave is always and everywhere transverse to the direction of propagation. We can solve the wave equation for any of the field components; for the parallel-plate case, it is most convenient to solve for E_y, from which all the other field components can be found using equations (Eqn. 18). Rewriting the wave equation (Eqn. 16) for they component, we have

$$\frac{\partial^2 E_y}{\partial x^2} + \overline{\gamma}^2 E_y - (j\beta)^2 E_y = 0 \qquad \text{(Eqn. 21)}$$

or

$$\frac{\partial^2 E_y}{\partial x^2} + h^2 E_y = 0 \qquad \text{(Eqn. 22)}$$

where $h^2 = \gamma^2 + (j\bar{\beta})^2 = \bar{\gamma}^2 + \omega^2\mu\epsilon$. We now express $E_y(x, y)$ as a product of two functions, one varying with x and another varying with z as

$$E_y(x,z) = E_y^0(x)e^{-\overline{\gamma}z}$$

Note that z variation of all the field components has the form $e^{-\overline{\gamma}z}$.

We can now rewrite (Eqn. 22) as an ordinary differential equation in ter1ns of $E_y^0(x)$:

$$\frac{d^2E_y^0(x)}{dx^2} + h^2E_y^0(x) = 0 \qquad \text{(Eqn. 23)}$$

Note that (Eqn. 23) is a second-order differential equation whose general solution is

$$E_y^0(x) = C_1\sin(hx) + C_2\cos(hx) \qquad \text{(Eqn. 24)}$$

where C_1, C_2, and h are constants to be determined by the boundary conditions. Note that the complete phasor for the y component of the electric field is

$$E_y(x,z) = [C_1\sin(hx) + C_2\cos(hx)]e^{-\overline{\gamma}z}$$

We now consider the boundary condition that requires the tangential electric field to be zero at the perfectly conducting plates. In other words,

$$\left.\begin{array}{l} E_y = 0 \text{ at } x = 0 \\ E_y = 0 \text{ at } x = a \end{array}\right\} \quad \text{for all } y \text{ and } z$$

Thus, $C_2 = 0$, and

$$E_y(x,z) = C_1\sin(hx)e^{-\overline{\gamma}z}$$

results from the first boundary condition. The second boundary condition (i.e., $Ey = 0$ at $x = a$) imposes a restriction on h; that is,

$$\boxed{h = \frac{m\pi}{a} \qquad m = 1, 2, 3, \ldots}$$

(Eqn. 25)

This result illustrates the particular importance of the constant h. Waveguide field solutions such as (24) can satisfy the necessary boundary conditions only for a discrete set of values of h. These values of h for which the field solutions of the wave equation can satisfy the boundary conditions are known as the *characteristic values* or *eigenvalues.*

The other wave field components can all be obtained from E_y using (18). We find $E_x = H_y = 0$, and

$$\text{Parallel-plate TE}_m, \; m = \pm 1, \pm 2, \ldots \quad \boxed{\begin{aligned} E_y &= C_1 \sin\left(\frac{m\pi}{a}x\right) e^{-\bar{\gamma} z} \\ H_z &= -\frac{1}{j\omega\mu}\frac{\partial E_y}{\partial x} = -\frac{m\pi}{j\omega\mu a} C_1 \cos\left(\frac{m\pi}{a}x\right) e^{-\bar{\gamma} z} \\ H_x &= -\frac{\bar{\gamma}}{j\omega\mu} E_y = -\frac{\bar{\gamma}}{j\omega\mu} C_1 \sin\left(\frac{m\pi}{a}x\right) e^{-\bar{\gamma} z} \end{aligned}} \qquad \text{(Eqn. 26)}$$

Note that $m = 0$ makes all fields vanish. Each integer value of m specifies a given field configuration, or *mode,* referred to in this case as the *TEm* mode. The configurations of fields for the cases of $m = 1$ and 2 are shown in Figure 3.

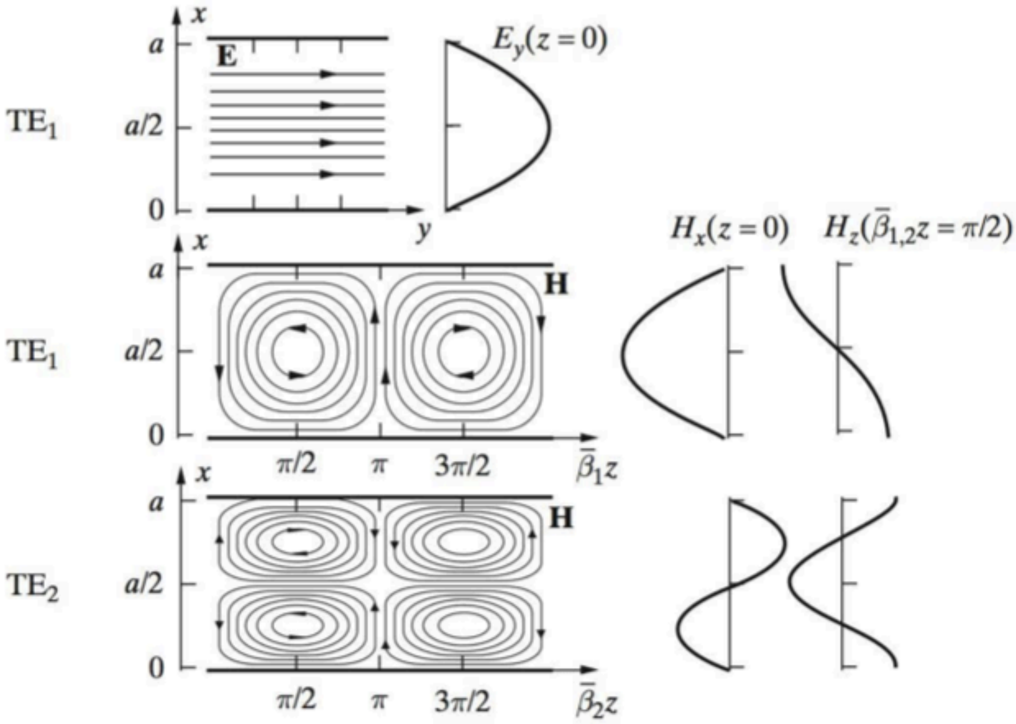


Figure 3 TE_1 and TE_2 modes.

The electric and magnetic field distributions for the TE_1 and the magnetic field distribution for the TE_2 modes in a parallel-plate waveguide.

Careful examination of the field structure for the TE_1 mode indicates that the magnetic field lines encircle the electric field lines (i.e., displacement current) in accordance with (10.la) and the right-hand rule. The same is true for the TE_2 mode, although the electric field distribution for this mode is not shown

Transverse Magnetic (TM) Waves.

For TM waves the magnetic field vector is always and everywhere transverse to the direction of propagation (i.e., $H_z = 0$), whereas the axial component of the electric field is nonzero (i.e., $H_z \neq 0$). We proceed with the solution in a similar manner as was done for TE waves, except that it is more convenient to use the wave equation for **H,** namely (10.5). The y component of this equation for H_y is in the form of (10.9) for E_y. Following the same procedure as for the TE case for E_y , the general solution for H_y is

$$H_y = [C_3 \sin(hx) + C_4 \cos(hx)]e^{-\bar{\gamma}z}$$

Since the boundary condition for the magnetic field is in ter1ns of its component nor1nal to the conductor boundary, it cannot be applied to H_y directly. However, using (10.6), we can write the tangential component of the electric field E_y in ter1ns of H_y as

$$E_z = \frac{1}{j\omega\epsilon}\frac{\partial H_y}{\partial x} = \frac{h}{j\omega\epsilon}[C_3 \cos(hx) - C_4 \sin(hx)]e^{-\bar{\gamma}z}$$

We now note that for all y and z:

$$E_z = 0 \quad \text{at } x = 0 \quad \rightarrow \quad C_3 = 0$$

and

$$E_z = 0 \quad \text{at } x = a \quad \rightarrow \quad \boxed{h = \frac{m\pi}{a} \qquad m = 0, 1, 2, 3, \ldots} \qquad \text{(Eqn. 27)}$$

The only other nonzero field component is E_x , which can be simply found from H_y or E_z . The nonzero field components for this mode, denoted as the TM_m mode, are then

$$\text{Parallel-plate TM}_m,\ m = 0, \pm1, \pm2, \ldots \quad \boxed{\begin{aligned} H_y &= C_4 \cos\left(\frac{m\pi}{a}x\right)e^{-\bar{\gamma}z} \\ E_x &= \frac{\bar{\gamma}}{j\omega\epsilon}H_y = \frac{\bar{\gamma}}{j\omega\epsilon}C_4 \cos\left(\frac{m\pi}{a}x\right)e^{-\bar{\gamma}z} \\ E_z &= \frac{jm\pi}{\omega\epsilon a}C_4 \sin\left(\frac{m\pi}{a}x\right)e^{-\bar{\gamma}z} \end{aligned}} \qquad \text{(Eqn. 28)}$$

The configurations of the fields for $m = 1$ and 2 (i.e., for the TM_1 and TM_2 modes) are shown in Figure 4.

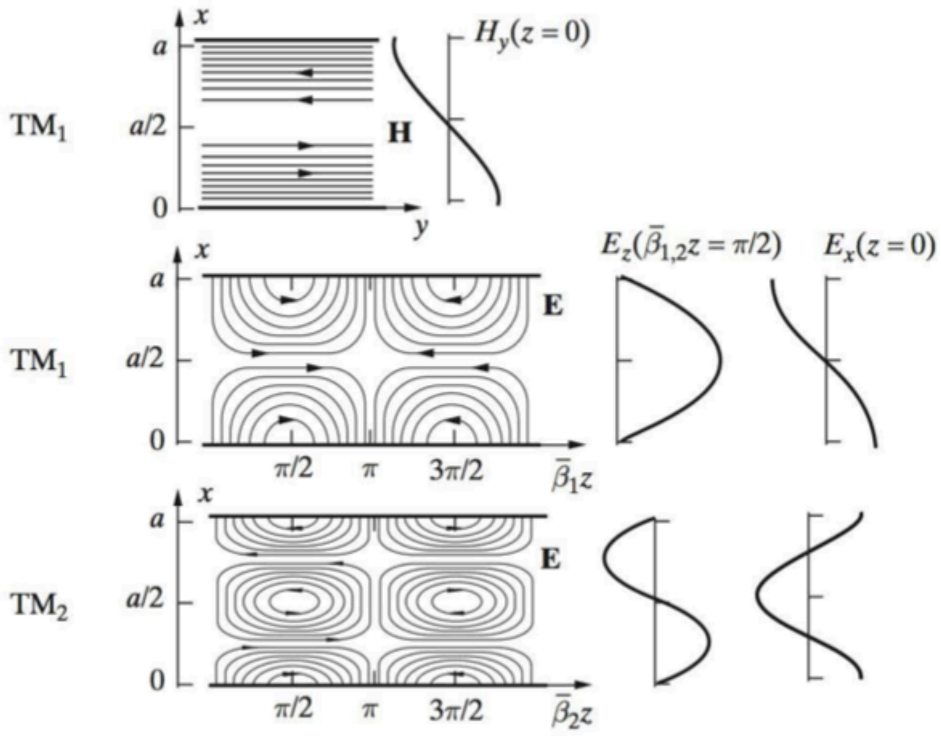


Figure 3 TE_1 and TE_2 modes.

Figure 4 TM_1 and TM_2 modes. The electric and magnetic field distributions for TM_1 and TM_2 modes in a parallel plane waveguide. Note that for TM_1, the electric field lines encircle the magnetic field lines (Faraday's law) in accordance with (13.1c). The same is true for TM2, although the magnetic field distribution for TM2 is not shown.

Transverse Electromagnetic (TEM) Waves.

Note that contrary to the TE case, the TM solutions do not all vanish for $m = 0$. Since E_z is zero for $m = 0$, the TM_o mode is actually a transverse electromagnetic (or TEM) wave. In this case, we have

$$\text{Parallel-plate TEM} \qquad \boxed{\begin{aligned} H_y &= C_4 e^{-\bar{\gamma}z} \\ E_x &= \frac{\bar{\gamma}}{j\omega\epsilon} C_4 e^{-\bar{\gamma}z} \\ E_z &= 0 \end{aligned}} \qquad \text{(Eqn. 29)}$$

For this case, we have $h^2 = 0$, so $\bar{y}^2 = j\bar{\beta} = j\bar{\beta}$ where $\omega\sqrt{\mu\epsilon}$ is the phase constant for a uniform plane wave in the unbounded lossless medium. The field configuration for the TM_o or TEM wave is shown in Figure 5 below.

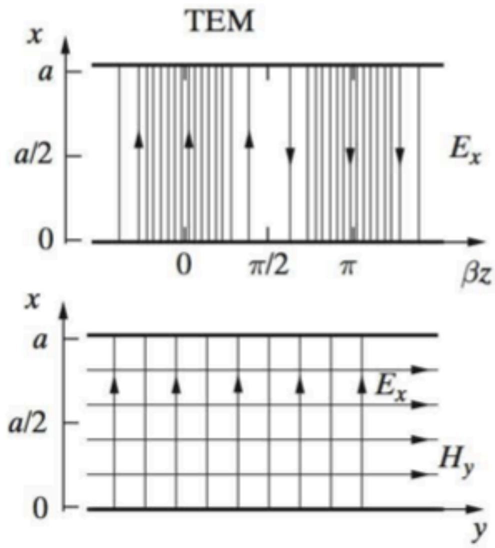


Figure 5: TEM mode. Electric and magnetic fields between parallel planes for the TEM (TMo) mode. Only electric field lines are shown in the top panel; the magnetic fields lines are out of (or into) the page

CUT-OFF FREQUENCY, PHASE VELOCITY, AND WAVELENGTH

Close examination of the solutions for TE and TM waves reveal a number of common characteristics: (i) $\boldsymbol{E}$ and $\boldsymbol{H}$ have sinusoidal standing-wave distributions in the x direction, (ii) any $x - y$ plane is an equiphase plane, or, in other words, the TE and TM waves are plane waves with surfaces of constant phase given by $z = const.$, and (iii) these equiphase surfaces progress (propagate) along the waveguide with a velocity $\bar{v}_p = \omega/\bar{\beta}$. To see this, consider any of the field components - say, E_y, for TE waves. For a propagating wave, with $\bar{\gamma} = j\bar{\beta}$, and assuming C_1 to be real, from (Eqn. 26), we have

$$\mathscr{E}_y(x, z, t) = C_1 \sin\left(\frac{m\pi}{a}x\right) \cos(\omega t - \bar{\beta}z)$$

To determine the conditions under which we can have a propagating wave (i.e., $\bar{\gamma}$ purely imaginary, that is, $\bar{\gamma} = j\bar{\beta}$, consider the definition of the eigenvalue parameter h:

$$h^2 = \bar{\gamma}^2 - (j\beta)^2 \quad \rightarrow \quad \bar{\gamma} = \sqrt{h^2 + (j\beta)^2} = \sqrt{\left(\frac{m\pi}{a}\right)^2 - \omega^2\mu\epsilon}$$

Note that this expression for h is valid for both TE and TM modes. Thus, the expressions given below for propagation constant, phase velocity, and wavelength are also equally valid for both TE and TM modes. From the preceding definition of h it is apparent that for each mode (TE or TM) identified by mode index m, there exists a critical or *cut-off* frequency f_{c_m} (or wavelength λ_{cm}) such that

$$\bar{\gamma} = 0 \quad \rightarrow \quad f_{c_m} = \frac{mv_p}{2a} = \frac{m}{2a\sqrt{\mu\epsilon}} \quad \text{or} \quad \lambda_{c_m} = \frac{v_p}{f_{c_m}} = \frac{2a}{m} \qquad \text{(Eqn. 29)}$$

where $v_p = 1/\sqrt{\mu\epsilon}$ is the *intrinsic* phase velocity for uniform plane waves in the unbounded lossless medium. Note that $v_p = (\mu\epsilon)^{-1/2}$ depends only on E and μ and is thus indeed an intrinsic property of the medium. For $f > f_{c_m}$, the propagation constant γ is purely imaginary and is given by

$$\boxed{\bar{\gamma} = j\bar{\beta}_m = j\sqrt{\omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2} = j\beta\sqrt{1 - \left(\frac{f_{c_m}}{f}\right)^2} \qquad f > f_{c_m}} \qquad \text{(Eqn. 30)}$$

where we now start to use $\bar{\beta}_m$ instead of $\bar{\beta}$ to underscore the dependence of the phase constant on the mode index m. Note that $\bar{\beta}_m$ is the phase constant (sometimes referred to as the longitudinal phase constant or the propagation constant) for mode m, and $\beta = \omega\sqrt{\mu\epsilon}$ is the intrinsic phase constant, being that for uniform plane waves in the unbounded lossless medium. For $f < f_{c_m}$, y is a real number, given by

$$\boxed{\bar{\gamma} = \bar{\alpha}_m = \sqrt{\left(\frac{m\pi}{a}\right)^2 - \omega^2\mu\epsilon} = \beta\sqrt{\left(\frac{f_{c_m}}{f}\right)^2 - 1} \qquad f < f_{c_m}}$$

where α_m is the attenuation constant, in which case the fields attenuate exponentially in the z direction, without any wave motion.

For example, the expression for the $\xi_y$component for TE waves with C_1 real for the case when $f < f_{c_m}$, is

$$\mathscr{E}_y(x, z, t) = C_1 e^{-\bar{\alpha} z} \sin\left(\frac{m\pi}{a}x\right)\cos(\omega t)$$

Such a non-propagating mode is called an evanescent wave. For $f > f_{c_m}$, the propagating wave has a wavelength $\lambda_m =$ (sometimes referred to as the guide wavelength) and phase velocity v_{p_m} given as

$$\boxed{\begin{aligned}\bar{\lambda}_m &= \frac{2\pi}{\bar{\beta}_m} = \frac{\lambda}{\sqrt{1-(f_{c_m}/f)^2}} \\ \bar{v}_{p_m} &= \frac{\omega}{\bar{\beta}_m} = \frac{v_p}{\sqrt{1-(f_{c_m}/f)^2}}\end{aligned}} \qquad \text{(Eqn. 31)}$$

where $\lambda = 2\pi/\beta = v_p/f$ is the intrinsic wavelength, being the wavelength for uniform plane waves in the unbounded lossless medium, and thus depending only on μ, ϵ, and frequency f. Note that for the TM_o mode (which is in fact a TEM wave, as as studied in READING ASSIGNMENT I), $\bar{v}_p$ is equal to the intrinsic phase velocity $\bar{v}_p = (\sqrt{\mu\epsilon})$ in the unbounded loss- less medium and is thus independent of frequency. Furthermore, the phase constant β_0 and wavelength 'λ_0 are also equal to their values for a uniform plane wave in an unbounded lossless medium. There is no cut-off frequency for this mode since the propagating field solutions are valid at any frequency.

For TE and TM waves in general, it is clear from this discussion that the wavelength $\bar{\lambda}_m$, as observed along the guide, is longer than the corresponding wavelength in an unbounded lossless medium by the factor $[1 - (f_{cm}/f)^2]^{-1/2}$. Also, the velocity of phase progression inside the guide is likewise greater than the corresponding intrinsic phase velocity. Note, however, that *Vp* is not the velocity with which energy or information propagates, so $\bar{v}_p >$ v_p > *vp* does not pose any particular dilemma.

WAVEGUIDES

Generally, if the frequency of a signal or a particular band of signals is high, the bandwidth utilization is high as the signal provides more space for other signals to get accumulated. However, high frequency signals can't travel longer distances without getting attenuated. It is obvious that transmission lines help the signals to travel longer distances. Microwaves propagate through microwave circuits, components, and devices, which act as a part of Microwave transmission lines, broadly called as Waveguides.

A hollow metallic tube of uniform cross-section for transmitting electromagnetic waves by successive reflections from the inner walls of the tube is called as a **Waveguide**

Types of Waveguides

There are five types of waveguides as shown and presented in the figure 6 below:

1. Rectangular waveguide
2. Circular waveguide
3. Elliptical waveguide
4. Single-ridged waveguide
5. Double-ridged waveguide

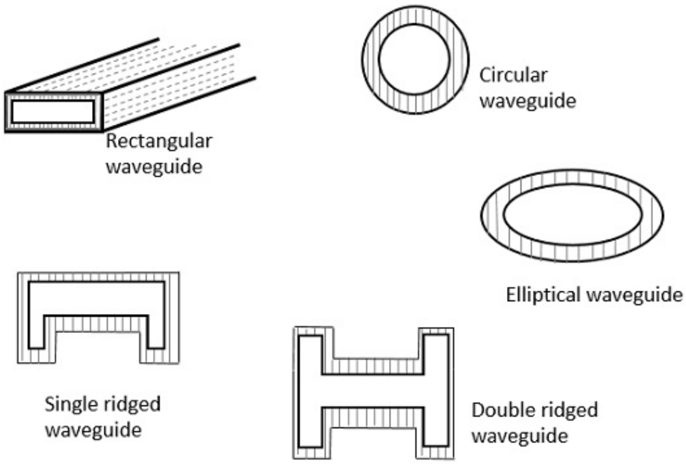


Figure 6: Types of waveguides

Waveguides, like transmission lines, are structures used to guide electromagnetic waves from point to point. However, the fundamental characteristics of waveguide and transmission line waves (modes) are quite different. The differences in these modes result from the basic differences in geometry for a transmission line and a waveguide.

Waveguides can be generally classified as either metal waveguides or dielectric waveguides. Metal waveguides normally take the form of an enclosed conducting metal pipe. The waves propagating inside the metal waveguide may be characterized by reflections from the conducting walls. The dielectric waveguide consists of dielectrics only and employs reflections from dielectric interfaces to propagate the electromagnetic wave along the waveguide.

RECTANGULAR WAVEGUIDE defined by Maxwell's Equations

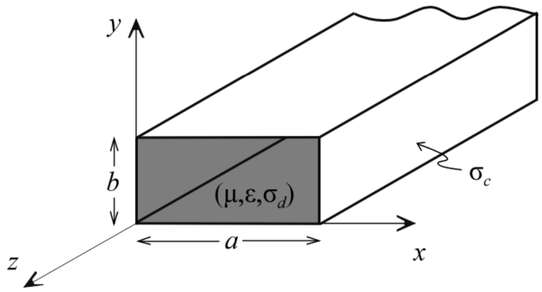


Figure 7: Rectangular waveguide

As presented earlier in the parallel plane wave guide Eqn. 13.a and 13.c earlier, given any time-harmonic source of electromagnetic radiation, the phasor electric and magnetic fields associated with the electromagnetic waves that propagate away from the source through a medium characterized by (μ, ε) must satisfy the source-free Maxwell's equations (in phasor form) given by

$$\nabla \times \tilde{\boldsymbol{E}} = -j\omega\mu\tilde{\boldsymbol{H}} \qquad \text{(Eqn. 32.a)}$$

$$\nabla \times \tilde{\boldsymbol{H}} = j\omega\varepsilon\tilde{\boldsymbol{E}} \qquad \text{(Eqn. 32.c)}$$

NOTE: Eqn. 13.a and Eqn. 32.a, as well as Eqn. 13.c and Eqn. 32.c are same… and so on.

The source-free Maxwell's equations can be manipulated into wave equations for the electric and magnetic fields (as was shown in the case of plane waves) which is very similar with the rectangular waveguide in Figure 7 above. These wave equations are

$$\nabla^2 \tilde{\boldsymbol{E}} + k^2 \tilde{\boldsymbol{E}} = 0 \qquad k = \omega\sqrt{\mu\varepsilon}$$

$$\nabla^2 \tilde{\boldsymbol{H}} + k^2 \tilde{\boldsymbol{H}} = 0$$

where the wave number k is real-valued for lossless media and complex- valued for lossy media. The electric and magnetic fields of a general wave propagating in the $+z-$ direction (either unguided, as in the case of a plane wave or guided, as in the case of a transmission line or waveguide) through an arbitrary medium with a propagation constant of ã are characterized by a $z-$ dependence of $e^{-\overline{\gamma}z}$. The electric and magnetic fields of the wave may be written in rectangular coordinates as

$$\tilde{\boldsymbol{E}}(x,y,z) = \tilde{\boldsymbol{e}}(x,y)\, e^{-\gamma z} \qquad \gamma = \alpha + j\beta$$
$$\tilde{\boldsymbol{H}}(x,y,z) = \tilde{\boldsymbol{h}}(x,y)\, e^{-\gamma z}$$

where α is the wave attenuation constant and β is the wave phase constant. The propagation constant is purely imaginary ($a = 0, \gamma = j\beta$) when the wave travels without attenuation (no losses) or complex-valued when losses are present.

The transverse vectors $[\tilde{\boldsymbol{e}}(x,y), \tilde{\boldsymbol{h}}(x,y)]$ in the general wave field expressions may contain both transverse field components and longitudinal field components. By expanding the curl operator of the source free Maxwell's equations in rectangular coordinates, we note that the derivatives of the transverse field components with respect to z are

$$\frac{\partial \tilde{E}_x}{\partial z} = -\gamma \tilde{E}_x \qquad \frac{\partial \tilde{E}_y}{\partial z} = -\gamma \tilde{E}_y$$
$$\frac{\partial \tilde{H}_x}{\partial z} = -\gamma \tilde{H}_x \qquad \frac{\partial \tilde{H}_y}{\partial z} = -\gamma \tilde{H}_y$$

If we equate the vector components on each side of the two Maxwell curl equations, we find

$$j\omega\varepsilon\tilde{E}_x = \frac{\partial\tilde{H}_z}{\partial y} + \gamma\tilde{H}_y \qquad \text{(Eqn. 33.a)}$$

$$j\omega\varepsilon\tilde{E}_y = -\gamma\tilde{H}_x - \frac{\partial\tilde{H}_z}{\partial x} \qquad \text{(Eqn. 33.b)}$$

$$j\omega\varepsilon\tilde{E}_z = \frac{\partial\tilde{H}_y}{\partial x} - \frac{\partial\tilde{H}_x}{\partial y} \qquad \text{(Eqn. 33.c)}$$

$$-j\omega\mu\tilde{H}_x = \frac{\partial\tilde{E}_z}{\partial y} + \gamma\tilde{E}_y \qquad \text{(Eqn. 34.a)}$$

$$-j\omega\mu\tilde{H}_y = -\gamma\tilde{E}_x - \frac{\partial\tilde{E}_z}{\partial x} \qquad \text{(Eqn. 34.b)}$$

$$-j\omega\mu\tilde{H}_z = \frac{\partial\tilde{E}_y}{\partial x} - \frac{\partial\tilde{E}_x}{\partial y} \qquad \text{(Eqn. 34.c)}$$

We may manipulate the first two equation above to solve for the longitudinal field components in terms of the transverse field components.

Solve (Eqn. 33.a) and (Eqn. 34.b) for $\tilde{H}_y$ $\quad \tilde{E}_x = \frac{1}{h^2}\left(-\gamma\frac{\partial\tilde{E}_z}{\partial x} - j\omega\mu\frac{\partial\tilde{H}_z}{\partial y}\right)$

Solve (Eqn. 33.b) and (Eqn. 34.a) for $\tilde{H}_x$ $\quad \tilde{E}_y = \frac{1}{h^2}\left(-\gamma\frac{\partial\tilde{E}_z}{\partial y} + j\omega\mu\frac{\partial\tilde{H}_z}{\partial x}\right)$

Solve (Eqn. 33.b) and (Eqn. 34.a) for $\tilde{E}_y$ $\quad \tilde{H}_x = \frac{1}{h^2}\left(j\omega\varepsilon\frac{\partial\tilde{E}_z}{\partial y} - \gamma\frac{\partial\tilde{H}_z}{\partial x}\right)$

Solve (Eqn. 33.a) and (Eqn. 34.b) for $\tilde{E}_x$ $\quad \tilde{H}_y = \frac{1}{h^2}\left(-j\omega\varepsilon\frac{\partial\tilde{E}_z}{\partial x} - \gamma\frac{\partial\tilde{H}_z}{\partial y}\right)$

where the constant h is defined by

$$h^2 = \gamma^2 + \omega^2\mu\varepsilon = \gamma^2 + k^2 \qquad \Rightarrow \qquad \gamma = \sqrt{h^2 - k^2}$$

The equations for the transverse fields in terms of the longitudinal fields describe the different types of possible modes for guided and unguided waves.

Transverse electromagnetic (TEM modes)	$\tilde{E}_z = 0,\ \tilde{H}_z = 0$	plane waves, transmission line modes
Transverse electric (TE modes)	$\tilde{E}_z = 0,\ \tilde{H}_z \neq 0$	waveguide modes
Transverse magnetic (TM modes)	$\tilde{E}_z \neq 0,\ \tilde{H}_z = 0$	waveguide modes
Hybrid (EH or HE modes)	$\tilde{E}_z \neq 0,\ \tilde{H}_z \neq 0$	waveguide modes

For simplicity, consider the case of guided or unguided waves propagating through an ideal (lossless) medium where k is real-valued. For TEM modes, the only way for the transverse fields to be non-zero with $\tilde{E}_z = \tilde{H}_z = 0$ is for $h = 0$, which yields

$$\gamma = \sqrt{-k^2} = jk = \alpha + j\beta \qquad \rightarrow \qquad \beta = k \qquad \text{(TEM modes)}$$

Thus, for unguided TEM waves (plane waves) moving through a lossless medium or guided TEM waves (waves on a transmission line) propagating on an ideal transmission line, we have $\gamma = jk = j\beta$.

For the waveguide modes (TE, TM or hybrid modes), h cannot be zero since this would yield unbounded results for the transverse fields. Thus, $\beta \neq k$ for waveguides and the waveguide propagation constant can be written as

$$\gamma = \sqrt{h^2 - k^2} = \sqrt{-k^2\left(1 - \frac{h^2}{k^2}\right)} = jk\sqrt{1 - \left(\frac{h}{k}\right)^2}$$

The propagation constant of a wave in a waveguide (TE or TM waves) has very different characteristics than the propagation constant for a wave on a transmission line (TEM waves). The ratio of h/k in the waveguide mode propagation constant equation can be written in terms of the cut-off frequency f_c for the given waveguide mode as follows.

$$\frac{h}{k} = \frac{h}{\omega\sqrt{\mu\varepsilon}} = \frac{h}{2\pi f\sqrt{\mu\varepsilon}} = \frac{f_c}{f}$$

$$f_c = \frac{h}{2\pi\sqrt{\mu\varepsilon}} \qquad \text{(waveguide cutoff frequency)}$$

The waveguide propagation constant in terms of the waveguide cut-off frequency is

$$\gamma = jk\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

CIRCULAR WAVEGUIDE defined by Maxwell's Equations

The same techniques used to analyze the ideal rectangular waveguide may be used to determine the modes that propagate within an ideal circular waveguide [$radius\ = a$, filled with dielectric ($\mu\varepsilon$)] The separation of variables technique yields solutions for the circular waveguide TE and TM propagating modes in terms of Bessel functions. The cut-off frequencies for the circular waveguide can be written in terms of the zeros associated with Bessel functions and derivatives of Bessel functions.

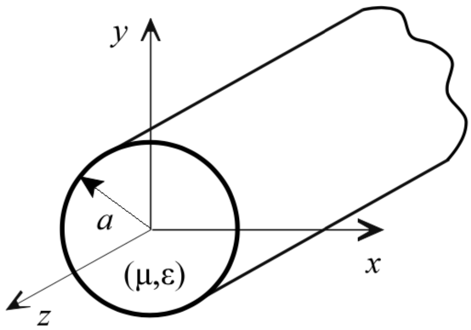


Figure 8: Circular waveguide

The cutoff frequencies of the TE and TM modes in a circular waveguide are given by

$$f_{c_{mn}}^{TE} = \frac{\chi'_{mn}}{2\pi a\sqrt{\mu\varepsilon}}$$

$$f_{c_{mn}}^{TM} = \frac{\chi_{mn}}{2\pi a\sqrt{\mu\varepsilon}}$$

where χ_{mn} and χ'_{mn} define the n^{th} zero of the m^{th}-order Bessel function and Bessel function derivative, respectively. The values of these zeros are shown in the tables below.

TE modes

χ'_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

TM modes

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Note that the dominant mode in a circular waveguide is the TE_{11} mode, followed in order by the TM_{01} mode, the TE_{21} mode and the TE_{01} mode.

Example (Circular waveguide)

Design an air-filled circular waveguide yielding a frequency separation of 1 GHz between the cut-off frequencies of the dominant mode and the next highest mode.

The cut-off frequencies of the TE_{11} mode (dominant mode) and the TM_{01} mode (next highest mode) for an air-filled circular waveguide are

$$f_{c_{11}}^{TE} = \frac{\chi'_{11}}{2\pi a\sqrt{\mu_o \varepsilon_o}} = \frac{1.8412}{2\pi a\sqrt{\mu_o \varepsilon_o}}$$

$$f_{c_{01}}^{TM} = \frac{\chi_{01}}{2\pi a\sqrt{\mu_o \varepsilon_o}} = \frac{2.4049}{2\pi a\sqrt{\mu_o \varepsilon_o}}$$

For a difference of 1 GHz between these frequencies, we write

$$f_{c_{01}}^{TM} - f_{c_{11}}^{TE} = \frac{2.4049 - 1.8412}{2\pi a\sqrt{\mu_o \varepsilon_o}} = 10^9$$

Solving this equation for the waveguide radius gives:

$$a = \frac{2.4049 - 1.8412}{2\pi 10^9 \sqrt{\mu_o \varepsilon_o}} = 2.6896 \text{ cm}$$

The corresponding cut-off frequencies for this waveguide are

$$f_{c_{11}}^{TE} = \frac{\chi'_{11}}{2\pi a \sqrt{\mu_o \varepsilon_o}} = \frac{1.8412}{2\pi(0.026896)\sqrt{\mu_o \varepsilon_o}} = 3.266 \text{ GHz}$$

$$f_{c_{01}}^{TM} = \frac{\chi_{01}}{2\pi a \sqrt{\mu_o \varepsilon_o}} = \frac{2.4049}{2\pi(0.026896)\sqrt{\mu_o \varepsilon_o}} = 4.266 \text{ GHz}$$

COMPARISON OF WAVEGUIDE AND TRANSMISSION LINE CHARACTERISTICS

	TRANSMISSION LINE	WAVEGUIDE
1.	Two or more conductors separated by some insulating medium (two-wire, coaxial, microstrip, etc.).	Metal waveguides are typically one enclosed conductor filled with an insulating medium (rectangular, circular) while a dielectric waveguide consists of multiple dielectrics.
2.	Normal operating mode is the TEM or quasi-TEM mode (can support TE and TM modes but these modes are typically undesirable)	Operating modes are TE or TM modes (cannot support a TEM mode).
3.	No cut-off frequency for the TEM mode. Transmission lines can transmit signals from DC up to high frequency.	Must operate the waveguide at a frequency above the respective TE or TM mode cut-off frequency for that mode to propagate.
4.	Significant signal attenuation at high frequencies due to conductor and dielectric losses.	Lower signal attenuation at high frequencies than transmission lines.
5.	Small cross-section transmission lines (like coaxial cables) can only transmit low power levels due to the relatively high fields concentrated at specific locations within the device (field levels are limited by dielectric breakdown).	Metal waveguides can transmit high power levels. The fields of the propagating wave are spread more uniformly over a larger cross-sectional area than the small cross-section transmission line.
6.	Large cross-section transmission lines (like power transmission lines) can transmit high power levels.	Large cross-section (low frequency) waveguides are impractical due to large size and high cost.

PRINCIPLE OF WAVEGUIDE COMPONENTS

Match Loads

Match loads is simply impedance matching and in general, this can be achieved by using a waveguide iris as shown in figure below.

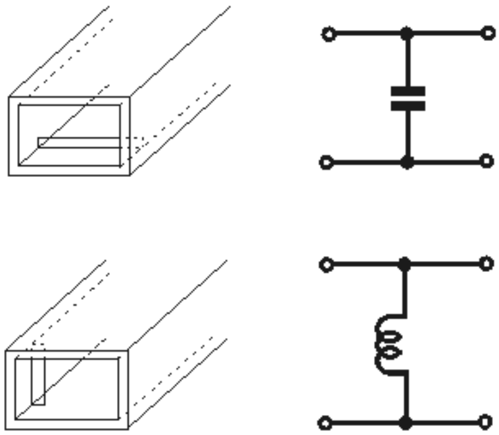


Figure 9: Impedance matching using a waveguide iris

The waveguide iris may either be on only one side of the waveguide, or there may be a waveguide iris on both sides to balance the system. A single waveguide iris is often referred to as an asymmetric waveguide iris or diaphragm and one where there are two, one either side is known as a symmetrical waveguide iris.

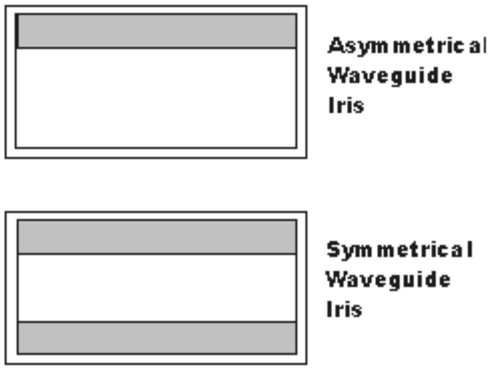


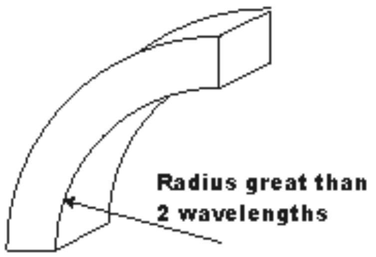
Figure 10: Symmetrical and asymmetrical waveguide iris implementations

A combination of both E and H plane waveguide irises can be used to provide both inductive and capacitive reactance. This forms a tuned circuit. At resonance, the iris acts as a high impedance shunt. Above or below resonance, the iris acts as a capacitive or inductive reactance.

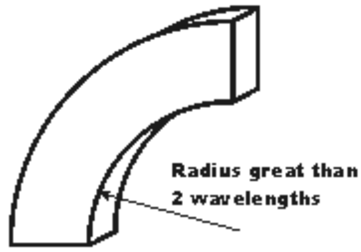
Waveguide Bends

Usually, a waveguide bends allows changes in the direction of the transmission and these bends are categorized into either waveguide E bend or H bend depending on the field it distorted. There are several ways in which waveguide bends can be accomplished which, may be used according to the applications and the requirements as listed and shown below.

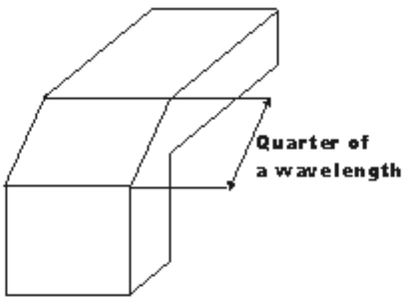
Waveguide E bend



Waveguide H bend



Waveguide sharp E bend



Waveguide sharp H bend

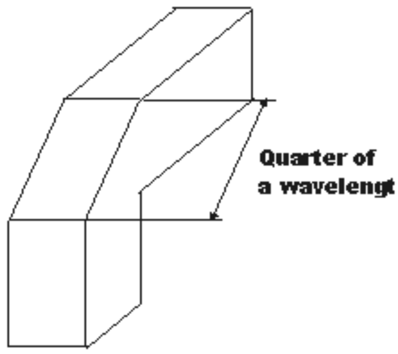


Figure 11: Waveguide BENDS

Each type of bend is achieved in a way that enables the signal to propagate correctly and with the minimum of disruption to the respective fields and hence to the overall signal. Both waveguide E or H bend are form of waveguide bend are very similar to another, except that the individual bend distorts the corresponding E or H field.

To prevent reflections in these waveguides, however, the corresponding (i.e., E and H) bends must have a radius greater than two wavelengths.

Attenuators

In order to control power levels in a microwave system by partially absorbing the transmitted microwave signal, attenuators are employed. Resistive films (dielectric glass slab coated with aquadag) are used in designing of both fixed and variable attenuators.

A co-axial fixed attenuator uses the dielectric lossy material inside the centre conductor of the co-axial line to absorb some of the centre conductor microwave power propagating through it dielectric rod decides the amount of attenuation introduced. The microwave power absorbed by the lossy material is dissipated as heat

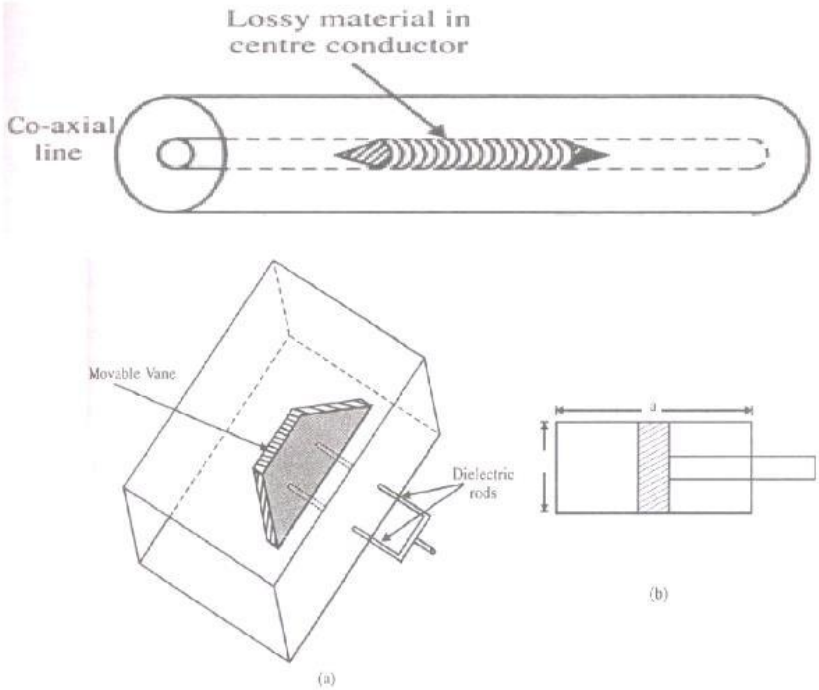
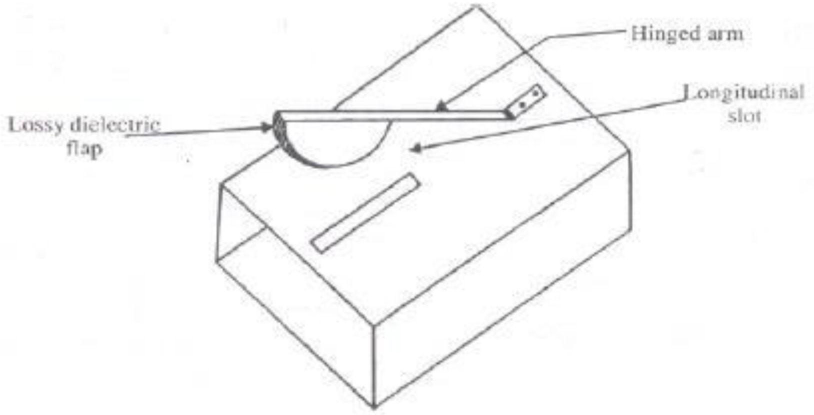


Figure 12: (a) Waveguide Attenuators

In waveguides, the dielectric slab coated with aduadag is placed at the centre of the waveguide parallel to the maximum E-field for dominant TE_{IO} mode. Induced current on the lossy material due to incoming microwave signal, results in power dissipation, leading to attenuation of the signal. The dielectric slab is tapered at both ends up to a length of more than half wavelength to reduce reflections as shown in figure 12. The dielectric slab may be made movable along the breadth of the waveguide by supporting it with two dielectric rods separated by an odd multiple of quarter guide wavelength and perpendicular to electric field.

Figure 12: (b) Waveguide Attenuators



Phase Shifters:

A microwave phase shifter is a two-port device which produces a variable shift in phase of the incoming microwave signal. A lossless dielectric slab when placed inside the rectangular waveguide produces a phase shift.

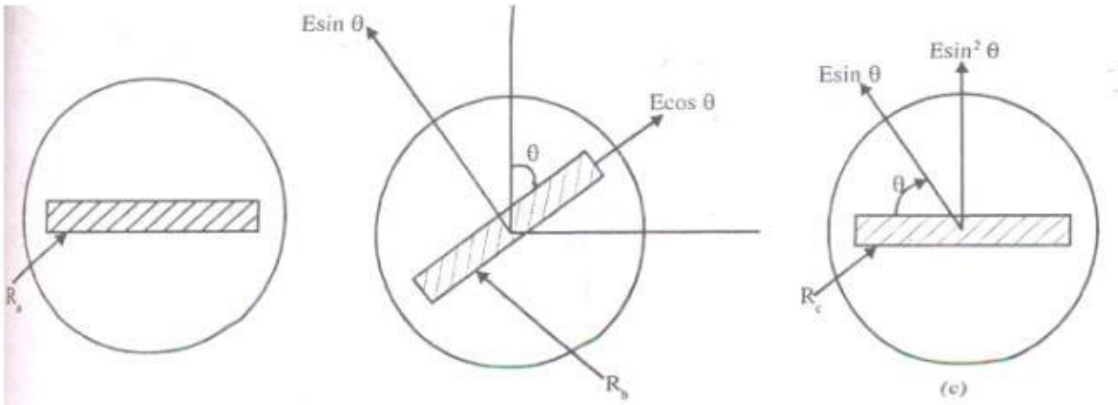
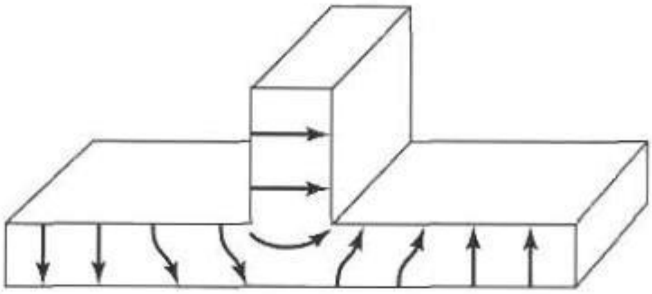


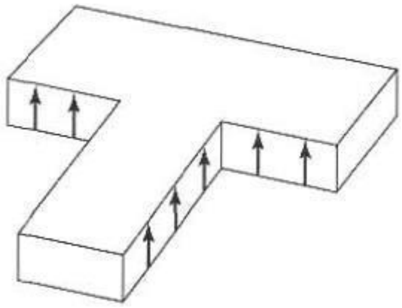
Figure 13: (b) Waveguide Phase Shifters

Multiport Junctions

T-Junction Power Divider: The T-junction power divider is a 3-port network that can be constructed either from a transmission line or from the waveguide depending upon the frequency of operation. For very high frequency, power divider using waveguide includes the following types: E-PLANE TEE, E-PLANE TEE, and E-H PLANE TEE or MAGIC TEE as presented in the figures below

E-PLANE TEE ***H-PLANE TEE:***





E-H PLANE TEE/MAGIC TEE

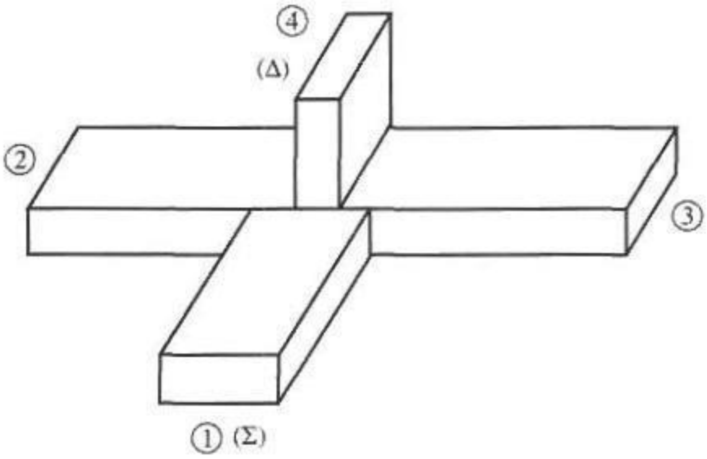


Figure 14: Waveguide Multiport junctions

Reading Assignment II

Read, understand, sketch, and label the following waveguide components:

i. **Couplers**

ii. **Isolators**

iii. **Duplexers and resonant cavities.**

MICROWAVE TUBES

Microwave tubes are a type of vacuum tube that generate and amplify high frequency microwave signals from 300 MHz to 300 GHz. They can generate high output power levels from a few hundred watts to more than 10 MW. Microwave tubes generally are classified into two major categories namely the linear beam tubes (O-Types) and the cross-field tubes (M-Types tubes). Microwave tubes are commonly used in Radio Detecting and Ranging (RaDaR) systems such as military radar, civilian radar-like weather detection, highway collision avoidance, electronic warfare, etc. As the most commonly used microwave tubes, classification of linear beam tubes is shown in the figure below.

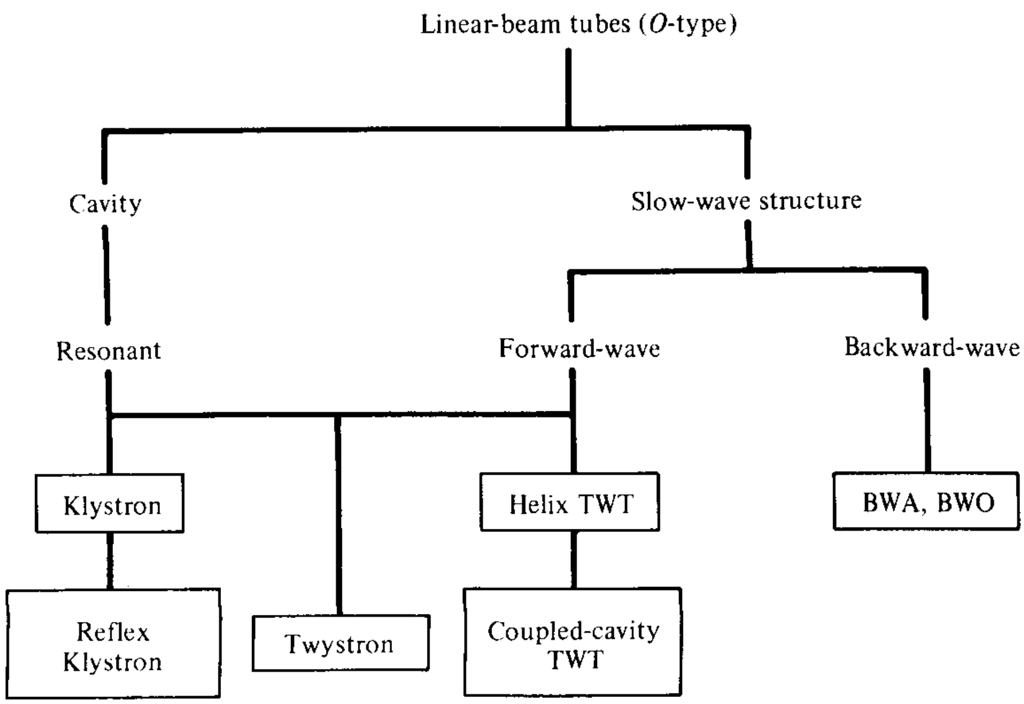


Figure 16: Classification of linear beam tubes.

Energy Transfer Mechanism

In most of the microwave tubes, the signal is placed in a cavity gap and electrons are forced to cross the gap at time when they face maximum opposition. Crossing the gap under opposition lead to transfer of energy to the cavity gap signal. When the gap voltage is sinusoidal time-varying and the charge crossing is continuous and uniform, which is usually the case, no net transfer of energy takes place between cavity and the charge crossing the gap. It is because the energy transfer is equal and opposite in direction during a half cycle when compared to half cycle resulting in no net transfer of energy in a cycle. To have net energy transfer, preferably maximum, from electron beam to gap signal voltage the distributed charge is compressed into a thin sheet or bunch, so that it requires less time to cross the gap and it is arranged such hat the bunch crossing is at peak gap voltage so that the bunch faces maximum opposition and retardation from the signal voltage.

When the gap voltage is sinusoidal and bunch crossing is at a uniform and constant rate, for maximum unidirectional flow of energy, there is only one instant, either at positive peak or negative peak, for the bunch to cross the gap. The bunch crossing hence must be once per cycle of the gap voltage. In case of bunch crossing at a uniform rate of f, transfer of maximum energy can take place only with a component of grid gap fields whose frequency is also f. Other components of the grid gap voltage like $2f, 4f, 8f$, etc., do not involve in the energy transfer, whereas the components $3f, 5f, 6f$, etc., and $f/2, f/3. f/4$, etc., the transfered amount of energy is negligible.

Klystron

Klystron is another type of linear-beam tube. In this case the electrons flow is in a straight line and a magnetic field is used to focalize the electron beam. They work on the principle of velocity modulation. Velocity modulation is the variation in the velocity of an electron beam when electrons in the beam are being accelerated and slowed down. This variation is normally achieved by varying the velocity of the beam, which causes the electrons to bunch up and generate some amount of RF current. Klystrons typically applies the transit-time effect principle by varying the velocity of an electron beam to form an electron bunch. The resonant cavities in the tube modulate the electric field along the axis. Depending on the number of the resonant cavities, klystrons are categorized into two-cavity klystrons, multi-cavity klystrons, and reflex klystrons.

Two Cavity Klystron

The two-cavity klystron is a widely used microwave amplifier operated by the principles of velocity and current modulation. It consists of an electron gun, focussing and accelerating grids, two identical cavities separated by a distance and at the far end a grounded collector plate. The electron gun emits electrons from the surface of its cathode and then they are focussed into a beam. Using a d.c accelerating positive voltage the beam is accelerated to high velocities.

Characteristics of two cavity klystron

Efficiency:	about 40%
Power output in CW mode:	up to 500 kW at 10 GHz
Power output in Pulsed mode:	up to 30 MW at 10 GHz
Power gain:	about 30 dB

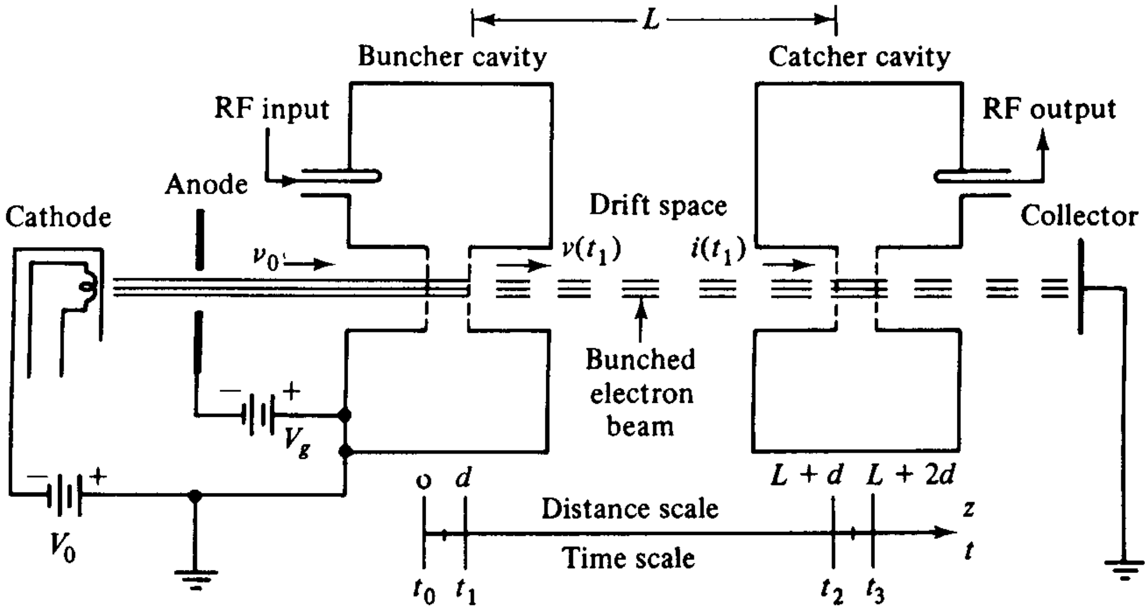


Figure 17: (a) The two-cavity klystron

Components of two cavity klystron

1. **Cathode:** Source of electrons
2. **Anode:** for formation of electron beam
3. **Bucher Cavity:** A reentrat type resonator cavity which is kept at a +ve voltage of Vo with respect to cathode to effect acceleration of electrons. RF input voltage of $V_1\ Sin\ \boldsymbol{\omega t}$ is applied to buncher cavity.
4. **Catcher cavity:** Similar to buncher cavity. The amplified output signal $V_2\ Sin\ \boldsymbol{\omega t}$ is obtained from this cavity.
5. **Collector:** The electrons after transfer of energy to catcher cavity are collected by the collector.

<u>**Definition of Parameters**</u>

The parameters used in the description and operation of two cavity klystron includes:

$V_0\ = DC$ voltage between cathode and buncher cavity.

$V_1\ =$ Amplitude of input RF signal, $V_1 << V_0$

$\omega\ =\ 2\pi f\ =$ Input signal angular frequency. It is also equal to resonant frequency of both the cavities.

v_o = Uniform velocity of electrons between cathode and buncher cavity.

t_o = Time at which electrons enter the buncher cavity t1 = Time at which electrons leave the buncher cavity

τ = Transit time of electrons in the buncher cavity = $t_1 - t_o$

θ_g = Angle /Phase variation of input signal during the transit time = ωT

β = Beam Coupling Coefficient of the buncher / Catcher Cavity

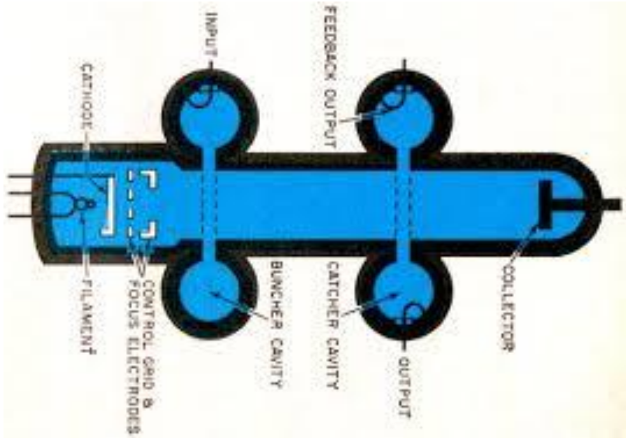


Figure 17: (b) The klystron

When the electrons enter the buncher cavity with uniform velocity v_o interact with the field due to input RF signal $V_1\ Sin\ \boldsymbol{\omega t}$ The time varying field in the cavity cause the electrons to accelerate or decelerate and there by electrons undergo velocity modulation.

Let $v(t_1)\ =$ Velocity of electrons at $t =\ t_1$at the output of buncher cavity

- This is a time varying quantity
- Refer fig-5.3 for velocity modulation of electrons

Let d = (cavity with of buncher) / (catcher cavity)

L = spacing between buncher and catcher cavities

NOTE THAT: L is the design parameter for optimum performance of the klystron amplifier

t_2= Time at which electrons enter the catcher cavity

t_3= Time at which electrons leave the catcher cavity

$$\tau \approx \frac{d}{v_o} = t_1 - \ t_o$$

Because $V_1 << V_0$

Evaluation of $\boldsymbol{v(t_1)}$

(t_1) is the instantaneous velocity of electrons which is a time varying quantity and primarily depends upon the average voltage in the gap during the time „τ ‟ i.e., during the time period (transit) the electrons are influenced by the field.

V_{aveg} = Average voltage in the gap during time τ

$$V_{avg} = \frac{1}{\tau}\int_{t_0}^{t_1} V_1 \sin\omega t \; dt$$

$$= \frac{-V_1}{\omega\tau}\left[\cos\omega t\right]_{t_0}^{t_1}$$

$$V_{avg} = \frac{-V_1}{\omega\tau}\left[\cos\omega t_1 - \cos\omega t_0\right] = V_{avg} = \frac{V_1}{\omega\tau}\left[\cos\omega t_o - \cos\omega t_1\right]$$

Where $\tau = t_1 - t_0$

$t_1 = t_0 + \tau = t_0 + \frac{d}{v_o}$

$$V_{avg} = \frac{V_1}{\omega\tau}\left[\cos\omega t_0 - \cos\left(\omega t_0 + \frac{\omega d}{v_0}\right)\right]$$

Where $\theta_g = \omega\tau = \frac{\omega d}{v_0}$ (5.3)

$$V_{avg} = \frac{V_1}{\omega\tau}\left[\cos\omega t_0 - \cos(\omega t_0 + \theta_g)\right]$$

Let $A = \omega t_0 + \frac{\theta_g}{2}$

and $B = \frac{\theta_g}{2}$

Since cos (A-B) – cos (A+B) = 2 sin A sin B

$$V_{avg} = V_1\left\{\frac{\sin(\omega d/2v_0)}{\omega d/2v_0}\right\} \sin\left(\omega t_0 + \frac{\omega d}{2v_0}\right)$$

$$= V_1 \frac{\sin\theta_g/2}{\theta_g/2}\sin\left(\omega t_0 + \frac{\theta_g}{2}\right)$$

β_i = beam coupling coefficient of input (buncher) cavity by definition

$$\beta_i = \frac{\sin\theta_g/2}{\theta_g/2} \qquad (5.4)$$

$$V_{avg} = V_1\beta_i \sin\left(\omega t_0 + \frac{\theta_g}{2}\right)$$

As we have seen earlier $v_0 = \sqrt{\frac{2eV_0}{m}}$

|||ly $\upsilon(t_1) = = \sqrt{\frac{2e}{m}}.\sqrt{V_1\beta_i\sin\left(\omega t_0 + \frac{\theta_g}{2}\right) + V_0}$

$$= \sqrt{\frac{2e}{m}V_0\left[1 + \frac{\beta_i V_1}{V_0}\sin\left(\omega t_0 + \frac{\theta_g}{2}\right)\right]}$$

Since $V_1 << V_0$, $\frac{\beta_i V_1}{V_0} << 1$

Using binomial expansion $\sqrt{1+x} = 1 + \frac{x}{2}$ *for* $x << 1$

$$v(t_1) = v_0\left[1 + \frac{\beta_i V_1}{2V_0}\sin\left(\omega t_0 + \frac{\theta_g}{2}\right)\right]$$

Since $\tau = t_1 - t_0$

$\theta_g = \omega\tau = \omega t_1 - \omega t_0$

$\omega t_0 = \omega t_1 + \omega t_0$

$\omega t_0 + \theta_g/2 = \omega t_1 - \theta_g + \theta_g/2 = \omega t_1 - \theta_g/2$

$$v(t_1) = v_0\left[1 + \frac{\beta_i V_1}{2V_0}\sin\left(wt_1 - \frac{\theta_g}{2}\right)\right] \tag{5.5}$$

$$v(t_1) = v_0\left[1 + \frac{\beta_i V_1}{2V_0}\sin\left(wt_0 + \frac{\theta_g}{2}\right)\right] \tag{5.6}$$

Numerical : *The parameters of a 2 cavity klystron are*

v_0 *= 1000 V,* I_0 *= 25 mA(beam current)*

d = 1mm, *f = 3 GHz*

Find out β_i *beam coupling coefficient*

Solution $v_0 = 0.593\times10^6\sqrt{V_0} = 0.593\times10^6\times\sqrt{1000}$

= 1.88×10^7 *m/s*

$$\theta_g = \frac{\omega d}{v_0} = \frac{2\pi\times3\times10^9\times10^{-3}}{1.88\times10^7} = 1.002\ rad$$

$$\beta_i = \frac{\sin\theta_g/2}{\theta_g/2} = \frac{\sin 0.5}{0.5} = 0.958$$

Reading Assign III

I. **Describe and further evaluate the related parameters associated with microwave tubes: Magnetron.**

II. **States any two of its applications**

Magnetron

The magnetron is cross field microwave tube. These tubes are normally circular in appearance. In magnetrons, the electric field and magnetic field run perpendicular to each other. The crossed electric and magnetic fields are used to generate high output power, which in turn is used in applications such as radar equipment. Magnetrons combine a vacuum tube with cavity resonators and a powerful magnet. A magnetron consists of an electron gun/cathode that emits the electrons and multiple anode cavities. A permanent magnet is placed on the backside of cathode and the space between the anode cavity and the cathode is called interacting space. There are two main types of magnetrons; pulsed magnetron and CW magnetron. Pulsed magnetrons are used in radars and CW ones are used in applications like microwave ovens.

MICROWAVE SEMI-CONDUCTOR DIODES

PN-junction Diodes: Semiconductor materials: Silicon-Germanium.

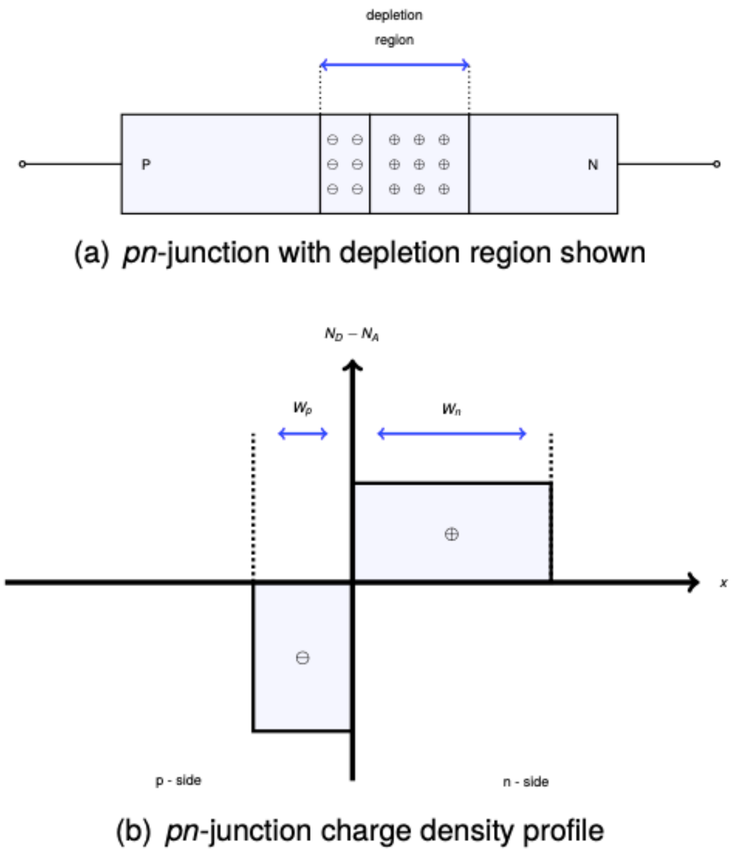


Figure 18: Simplified representation of a *pn*-junction

- When two semiconductor materials are brought into contact, the Fermi levels must become aligned so as to be the same throughout the crystal.
- Surplus electrons from the *n*-region will diffuse into the *p*-region leaving a region of net positive charge in the *n*-region, near the junction. Similarly, surplus holes from the *p*-region will diffuse into the *n*-region leaving a region of net negative charge in the *p*-region, near the junction.
- The region immediately either side of the junction will now have been depleted of majority carriers, and is therefore referred to as the *depletion region*, also referred to as the *space charge* region, as illustrated in figure 3.
- With no external applied voltage, a potential will exist across the *pn*-junction due to the build-up of carriers on each side. This is known as the *built-in* potential.
- When a forward bias is applied externally, the depletion region shrinks as negative charge carriers are repelled from the negative terminal towards the junction and holes are repelled from the positive terminal towards the junction. This reduces the energy required for charge carriers to cross the depletion region. As the applied voltage increases, current starts to flow across the junction once the applied voltage reaches the *Barrier potential*

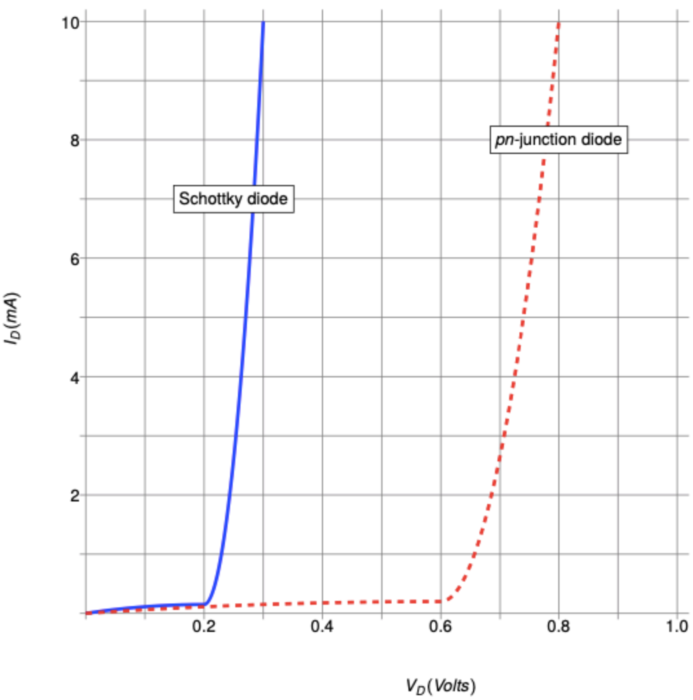


Figure 19: Schottky diode vs *pn*-junction forward bias characteristics (Silicon

Schottky Diode

- Schottky Diodes employ a Metal-Semiconductor Contact Junction. There is no *pn*-junction, as such.
- Schottky Diodes have very fast switching times due to their small capacitance and the fact that they are *majority carrier* devices.
- Schottky diodes have a very short reverse *recovery time*. For *pn*-junctions, the reverse recovery time is between 5 to 100 nS. For a Schottky diode it is normally between 5 and 100 nS.
- Schottky diodes are widely used in RF circuits as mixers and detectors.

Gunn diode

- Simply a slab of N-type GaAs (Gallium Arsenide) with no PN junctions
- The band structure of certain compound semiconductors, such as GaAs and InP, has two local minima in the conduction band: one where the electrons have a low effective mass and a high mobility and a second local minimum at a higher energy level where electrons have a higher effective mass and a lower mobility.
- This causes a concentration of free electrons called a domain which moves through the device from Cathode to Anode.

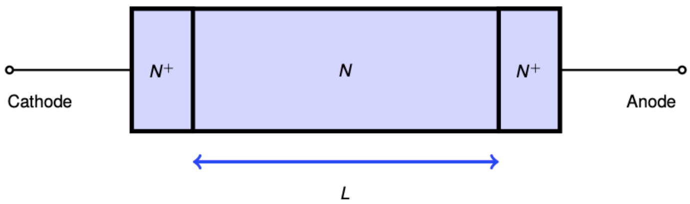


Figure 20: Gunn diode construction

Gunn Diode Characteristic

The drop in electron mobility with increasing electric field means that a sample of this material will exhibit a decrease in current with increasing applied voltage, that is to say a negative differential resistance.

At higher voltages, the normal increase of current with voltage relation resumes once the bulk of the carriers are kicked into the higher energy-mass valley.

Therefore, the negative resistance only occurs over a limited range of voltages, as illustrated.

OPTICAL FIBER

An optical fiber is a cylindrical dielectric waveguide made of low-loss materials such as silica glass. It has a central core in which the light is guided, embedded in an outer cladding of slightly lower refractive index in figure below. Light rays' incident on the core-cladding boundary at angles greater than the critical angle undergo total internal reflection and are guided through the core without refraction. Rays of greater inclination to the fiber axis lose part of their power into the cladding at each reflection and are not guided.

Fiber optic principles are essentially the same as those that apply in planar dielectric waveguides, except for the cylindrical geometry. In both types of waveguide light propagates in the form of modes. Each mode travels along the axis of the waveguide with a distinct propagation constant and group velocity, maintaining its transverse spatial distribution and its polarization.

One of the difficulties associated with light propagation in multimode fibers arises from the differences among the group velocities of the modes. This results in a variety of travel times so that light pulses are broadened as they travel through the fiber. This effect, called modal dispersion, limits the speed at which adjacent pulses can be sent without overlapping and therefore the speed at which a fiber-optic communication system can operate.

Modal dispersion can be reduced by grading the refractive index of the fiber core from a maximum value at its center to a minimum value at the core-cladding boundary. The fiber is then called a graded-index fiber, whereas conventional fibers

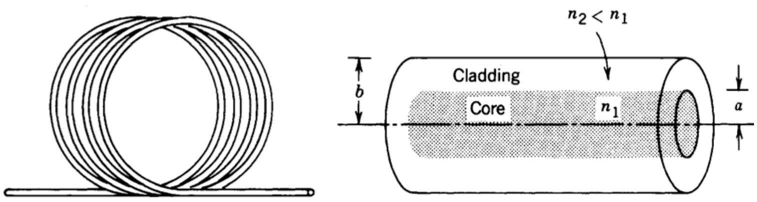


Figure 21: An optical fiber.

Structure and Physics of an Optical Fiber

Generally, an optical fibers used in communications have a very simple structure. They consist of two sections: the glass core and the cladding layer as shown in figure 22 below. The core is a cylindrical structure, and the cladding is a cylinder without a core. Core and cladding have different refractive indices, with the core having a refractive index, n_1, which is slightly higher than that of the cladding, n_2. It is this difference in refractive indices that enables the fiber to guide the light. Because of this guiding property, the fiber is also referred to as an "optical waveguide." As a minimum there is also a further layer known as the secondary cladding that does not participate in the propagation but gives the fiber a minimum level of protection. This second layer is referred to as a coating.

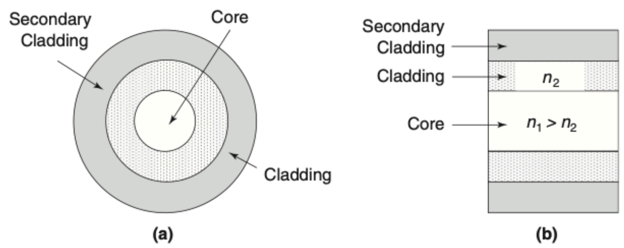
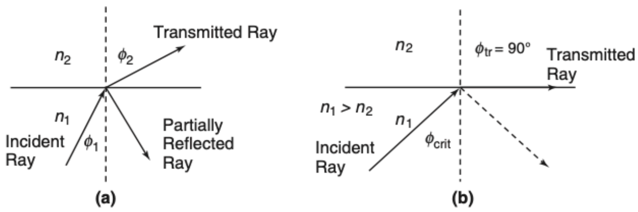


Figure 22: (a) Cross section and (b) longitudinal cross section of a typical optical fiber.



(b) Snell's law

The basics of light propagation can be discussed with the use of geometric optics. The basic law of light guidance is Snell's law. Consider two dielectric media with different refractive indices and with $n_1 > n_2$and that are in perfect contact, as shown in Snell's law Figure above. At the interface between the two dielectrics, the incident and refracted rays satisfy Snell's law of refraction — i.e.,

$$n_1 Sin\ \emptyset_1 = n_2\ Sin\ \emptyset_2$$

$$\frac{n_2}{n_1} = \frac{Sin\ \emptyset_1}{Sin\ \emptyset_2}$$

In addition to the refracted ray there is a small amount of reflected light in the medium with refractive index $n1$. Because $n_1 > n_2$then always $f_1 > f_2$As the angle of the incident ray increases there is an angle at which the refracted ray emerges parallel to the interface between the two dielectrics Figure 22 (b).

This angle is referred to as the critical angle, f_{crit} and from Snell's law is given by

$$Sin\ \emptyset_{crit} = \frac{n_2}{n_1}$$

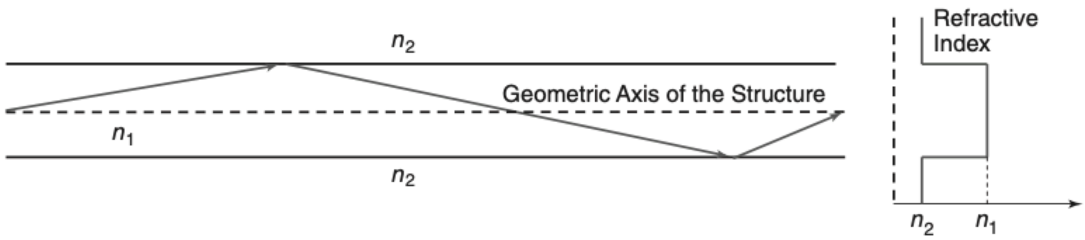


Figure 23: Light guidance using Snell's law.

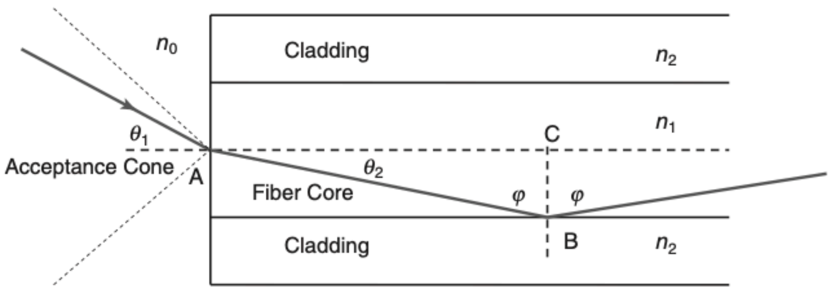


Figure 24: Geometry for the derivation of the acceptance angle.

If the angle of the incident ray is greater than the critical angle, the ray is reflected back into the medium with refractive index $n1$. This basic idea can be used to propagate a light ray in a structure with $n_1 > n_2$, and Figure 23 illustrates this idea. The light ray incident at an angle greater than the critical angle can propagate down the waveguide through a series of total reflections at the interface between the two dielectrics.

Figure 24 illustrates the geometry for the derivation of the acceptance angle. To satisfy the condition for total internal reflection, the ray arriving at the interface, between the fiber and outside medium, say air, must have an angle of incidence less than θ_{acc} otherwise the internal angle will not satisfy the condition for total reflection, and the energy of the ray will be lost in the cladding.

Consider that a ray with an incident angle less than the θ_{acc}, say q1, enters the fiber at the interface of the core (n_1) and the outside medium, say air (n0), and the ray lies in the meridional plane. From Snell's law at the interface, we obtain

$$n_o \, sin\theta_1 \; = \; n_1 \, sin\theta_2$$

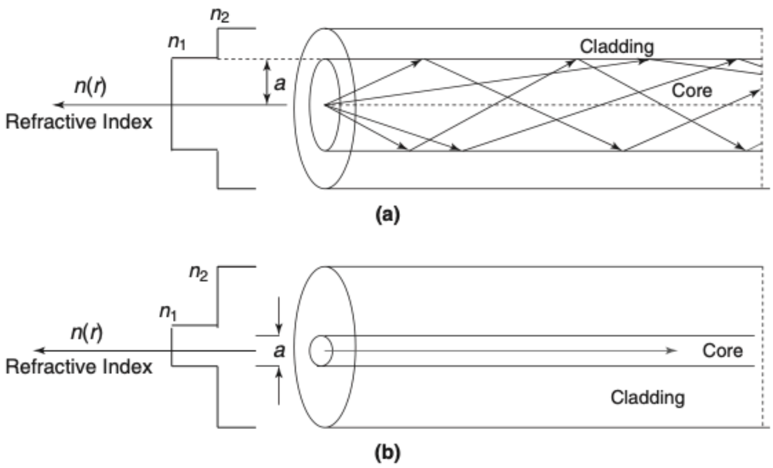


Fig. 25 (a) Multimode and (b) single-mode propagation for step index fiber.

HND II: Electrical & Electronics Engineering
Course CODE: EEE 445
Course TITLE: DIGITAL COMMUNICATION III (Data Communications)

Department of Electrical & Electronics
School of Engineering, The Federal Polytechnic Mubi PMB 35
Adamawa State

Course Lecturers:
Dr. A.liyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE
Mr. Mohammed Bobboi

LETURE NOTES 2: **RADAR**
RADAR EQUATION
RADAR FREQUENCIES
RADAR APPLICATIONS
DOPPLER RADAR SYSTEM
RADAR PROBLEMS

RADAR

The word *Radar* is an Acronym for RAdio Detection And Ranging.

Radar is a detection system that uses radio waves to determine the range, angle, or velocity of objects. It can be used to detect aircraft, ships, spacecraft, guided missiles, motor vehicles, weather formations, and terrain.

A radar system consists of a transmitter producing electromagnetic waves in the radio or microwaves domain, a transmitting antenna, a receiving antenna (often the same antenna is used for transmitting and receiving) and a receiver and processor to determine properties of the object(s). there are two basic radar systems

1. Monostatic radar systems
2. Bistatic radar systems

The system is called monostatic Radar because the system uses a single transceiver i.e., same antenna is used for transmission and reception. There are two sections of radar: The transmitter section and the receiver section.

Functions of Radar

- Detects the presence of target
- Gives the range of the target from the Radar station
- Gives the azimuth angle and elevation angle of the target
- Gives the radial velocity of target.

Radar block diagram

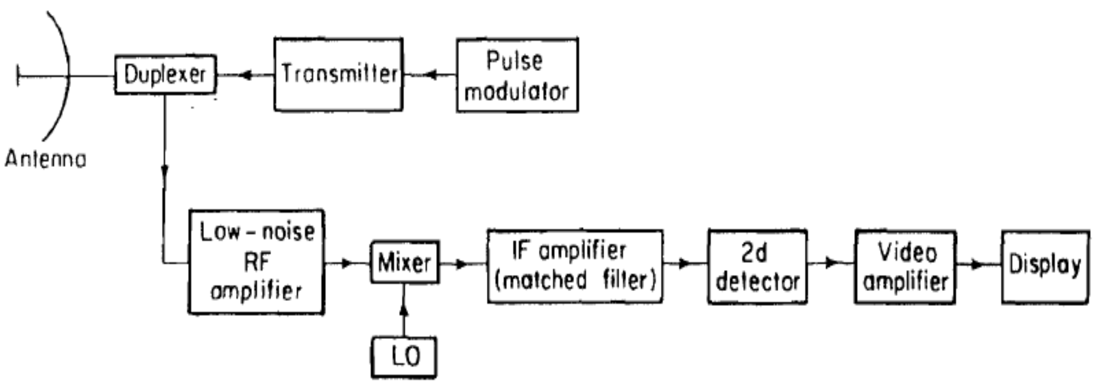


Figure 26: the block diagram of a monostatic Radar.

Transmitter section

- **Transmitter:** the transmitter may be a power amplifier such as klystron, travelling wave tube or transistor amplifier. This will generate the Electrical energy at R.F. (Radio Frequency).
- **Pulse modulator:** The power amplifier (Such as Klystron, TWT) produces a high-power signal, may be in terms of megawatts. Pulse modulator shown in the block is used as a switch, which will turn on and off the power amplifier.
- **Wave form generator**: A low power signal is produced by the waveform generator which is given as an input to the power amplifier.
- **Duplexer:** The duplexer allows a single antenna to be used on a time shared basis for both transmitting and receiving. The duplexer is generally a gaseous device that produces a short circuit at the input to the receiver when the transmitter is operating, so that high power flows to the antenna and not to the receiver. On the reception, the duplexer directs echo signal to the receiver and not to the transmitter. Solid state ferrite circulators and receiver protector devices can also be part of the duplexer

Receiver section:

- **Low noise RF amplifier:** The receiver is almost always a super heterodyne. LNA is used immediately after the antenna. This reduces the Noise Figures and produces the RF pulse proportional to the transmitted signal.
- **Mixer and local oscillator**: It convert the RF signal to an intermediated frequency where it is amplified by the IF amplifier. The IF frequency might be 30 or 60 MHz
- **IF amplifier**:
    i. It amplifies the IF pulse.
    ii. IF amplifier is designed as a matched filter which maximizes the output peak signal to mean noise ratio.
    iii. The matched filter maximizes the detectability of weak echo signals and attenuates unwanted signals.
    iv. The signal bandwidth of super heterodyne receiver is determined by the bandwidth of its IF stage.

v. For example when pulse width is of the order of 1µs the IF bandwidth would be about 1MHz.

- **Second Detector**: the IF amplifier followed by a crystal diode which is called the second detector or demodulator. Its purpose is to assist in extracting the echo signal modulation from the carrier. It is called as 2ndDetector since it is the second diode used in the chain. The first diode is used in the mixer. Output of the 2ndDetector is the Video Pulse.
- **Video amplifier**: It is designed to provide the sufficient amplification to rise the level of the input signal to a magnitude where it can be diplay (CRT or Digital computer).
- **Threshold decision**: The output of video amplifier is given to the threshold detector where it is decided whether the received signal is from a target or just because of the presence of noise.
- **Display**: The Display is generally a CRT (Cathode Ray Tube) (a) 'A' scope (b) PPI

i) 'A' scope provided Range and Echo power.

ii) PPI measures Range and bearing (azimuth angles)

iii) In addition there are other displays like 'B' scope, 'D' scope etc.

General Working Principle

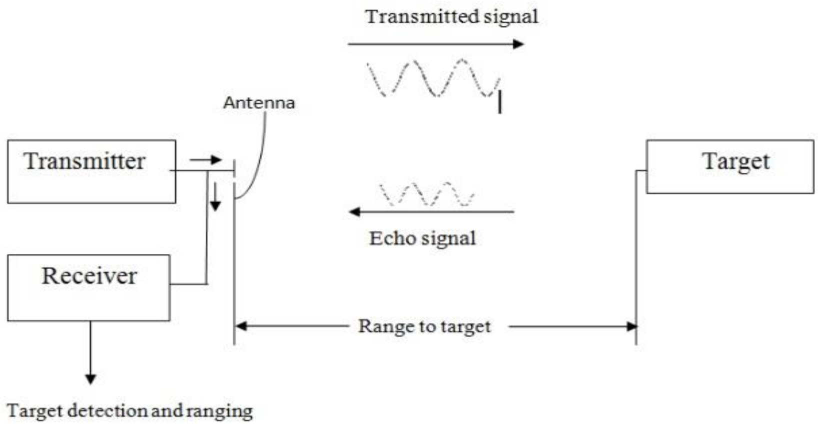


Fig. 27 Working principle of a RADAR system

RADAR System Definition of TERMS

When measuring a RADAR system Range, some terms/acronyms defined below are commonly used:

$\boldsymbol{PRF}$: The number of radar pulses transmitted per second is known as pulse repetition frequency or pulse repetition rate F_p.

$\boldsymbol{PRT}$: The time from beginning of first pulse to the beginning of the next is called pulse repetition time T_p.

$$F_p = \frac{1}{T_{p.}}$$

$\boldsymbol{T_R}$ Time taken by EM pulse to travel to target and come back to same antenna

R Range of target

$\boldsymbol{C}$ Velocity of EM waves = 3 x 10^8 Meters/sec

Rest Time or Receiver Time:

The time between two successive transmitted pulse is called as Rest Time or Receiver Time

Radar Range Determination

- The most common radar waveform is a train of narrow, rectangular-shape pulses modulating a sine wave carrier.
- The distance, or range, to the target is determined by measuring the time $\boldsymbol{T_R}$ taken by the pulse to travel to the target and return.
- Electromagnetic energy in free space travels with the speed of light $\boldsymbol{c}$ ***(****i.e. 3 x 10^8* $\boldsymbol{m/s)}$ therefore range $\boldsymbol{R}$ is given by : $\boldsymbol{R = c \times T_R / 2}$
- The range **R** in kilometres or nautical miles, and $\boldsymbol{T_R}$ in microseconds, the above relation becomes: $\boldsymbol{R(km) = 0.15 \times T_R}\ (\mu S)$ **or** $\boldsymbol{R(nmi) = 0.081 \times T_R}\ (\mu S)$
- Each microsecond of round-trip travel time corresponds to 0.081 nautical mile, **0.093** statute mile, **150** meters, **164** yards, or **492** feet.
- (1 mile = 0.8689 nautical mile or 1.6 km 1nauticalmile= 1.15078 miles or 1.8412km)
- It takes 12.35 µs for radar signal to travel a nautical mile and back

RADAR FREQUENCIES

- RF spectrum is very scarce and as such Radars are allotted only a certain frequency bands for their operation by International Telecom Union ITU
- During 2^{nd} world war, to keep the secrecy, certain code words were used. The same designations are continued even today
- Lema Band (L) 1GHZ-2GHZ, Sierra band(S) 2GHZ-4GHZ, Charlie Band (C) 4GHZ-8GHZ, Xera Band (X) 8GHZ-12GHZ

- ITU (International Telecommunication Union) allocated a portion of these bands for Radar

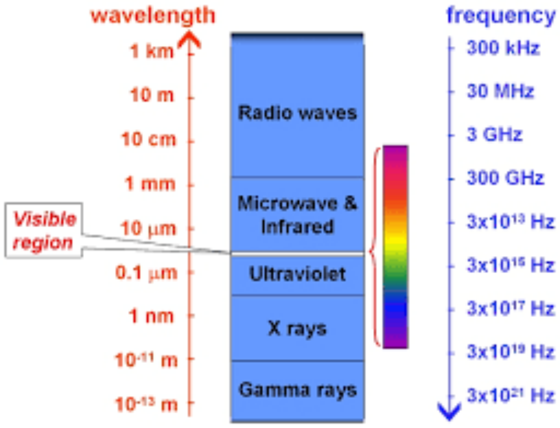


Fig. 29 Electromagnetic spectrum Wavelength and corresponding frequencies.

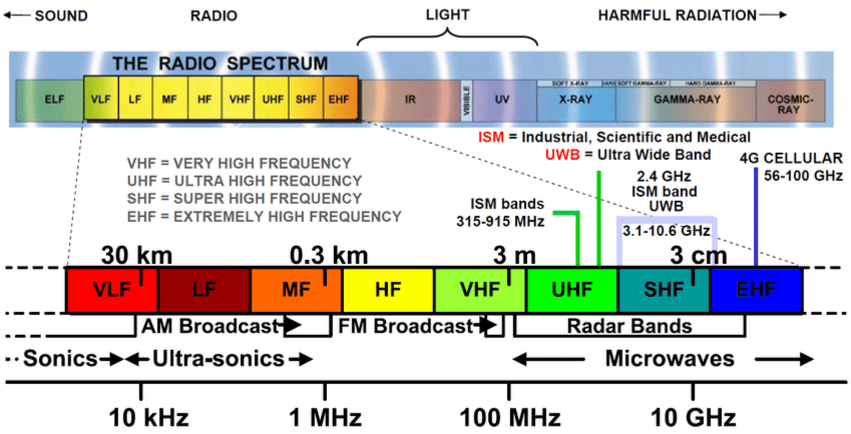


Fig. 29 (a) Electromagnetic Radio spectrum

Band designation	Nominal frequency range	Specific radiolocation (radar) bands based on ITU assignments for region 2
HF	3-30 MHz	
VHF	30-300 MHz	138-144 MHz 216-225
UHF	300-1000 MHz	420-450 MHz 890-942
L	1000-2000 MHz	1215-1400 MHz
S	2000-4000 MHz	2300-2500 MHz 2700-3700
C	4000-8000 MHz	5250-5925 MHz
X	8000-12,000 MHz	8500-10,680 MHz
K_u	12.0-18 GHz	13.4-14.0 GHz 15.7-17.7
K	18-27 GHz	24.05-24.25 GHz
K_a	27-40 GHz	33.4-36.0 GHz
mm	40-300 GHz	

Fig. 29 (c) Electromagnetic spectrum allocations for RADAR based on ITU

FIELDS OF APPLICATION OF RADAR SYSTEM

The following includes the application of a RADAR

a. Military
b. Remote Sensing
c. Air Traffic Control
d. Law Enforcement and Highway
e. Security
f. Aircraft Safety and Navigation
g. Ship Safety
h. Space
i. Miscellaneous Applications

Military:

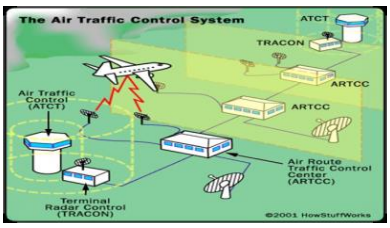
- Important part of air defence system, operation of offensive missiles & other weapons.
- Target detection, target tracking & weapon control.
- Also used in area, ground & air surveillance.

Air Traffic Control

- Used to safely control air traffic in the vicinity of the airports and enroute.
- Ground vehicular traffic & aircraft taxing.
- Mapping of regions of rain in the vicinity of airports & weather.

Law Enforcement & Highway Safety:

- Radar speed meters are used by police for enforcing speed limits.
- It is used for warning of pending collision, actuating air bag or warning of obstruction or people behind a vehicle or in the side blind zone



Remote Sensing

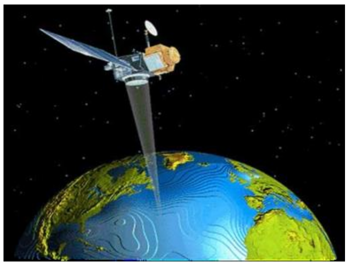
- Weather observation-
- T.V. Reporting
- Planetary observation
- Below ground probing
- Mapping of sea ice

Aircraft Safety & Navigation

- Low flying military aircrafts rely on terrain avoidance & terrain following radars to avoid collision with high terrain & obstructions

Ship Safety

- Radar is found on ships & boats for collision avoidance & to observe navigation buoys, when the visibility is poor.
- Shore based radars are used for surveillance of harbours & river traffic.



Space

- Space vehicles have used radar for clocking & for landing on the moon.
- Used for planetary exploration.
- Ground based radars are used for detection & tracking of satellites & other space objects.
- Used for radio astronomy.

Other APPLICATIONS

- It is used for in industry for the non-contact measurement of speed & distance.
- Used for oil & gas exploration.
- Used to study movements of insects & birds.

Prediction Of Range Performance

The simple form of the radar equation derived earlier expresses the maximum radar range $\boldsymbol{R_{max}}$ in terms of radar and target parameters:

$$R_{max} = \left[\frac{P_t \times G \times A_e \times \sigma}{(4\pi)^2 \times S_{min}}\right]^{1/4}$$

where

$\boldsymbol{P_t}$= transmitted power, measured in watts

$\boldsymbol{G}$ = antenna gain

$\boldsymbol{A_e}$ = antenna effective aperture, measured in **m²**

$\boldsymbol{\sigma}$ = radar cross section, measured in m^2

$\boldsymbol{S_{min}}$ = minimum detectable signal, measured in watts

All the parameters are to some extent under the control of the radar designer, except for the target cross section **σ.**

Doppler Effect

- Doppler effect implies that the frequency of a wave when transmitted by the source is not necessarily the same as the frequency of the transmitted wave when picked by the receiver.
- The received frequency depends upon the relative motion between the transmitter and receiver.
- If transmitter and receiver both are moving towards each other the received frequency higher, this is true even one is moving.
- If they are moving apart the received signal frequency decreases and if both are stationary, the frequency remains the same. This change in frequency is known as Doppler shift.

- Doppler shift depends upon the relative velocity between radar and target

RADAR EQUATION AND PROBLEMS

Reading Assignment IV

HND II: Electrical & Electronics Engineering
Course CODE: EEE 445
Course TITLE: DIGITAL COMMUNICATION III (Data Communications)

Department of Electrical & Electronics
School of Engineering, The Federal Polytechnic Mubi PMB 35 Adamawa State

Course Lecturers:
Dr. A.liyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE
Mr. Mohammed Bobboi

LETURE NOTES 3: **SATELLITES COMMUNICATIONS**
PRINCIPLE OF OPERATION
INTEL SAT SYSTEM
SATELLITE ORBIT
ADVANTAGE & DISADVANTAGE
APPLICATIONS
PROBLEM & SOLUTIONS
FREQUENCY SPECTRUM

INTRODUCTION SATELLITES SYSTEM

- Satellites are specifically made for telecommunication purpose. They are used for mobile applications such as communication to ships, vehicles, planes, hand-held terminals and for TV and radio broadcasting.
- They are responsible for providing these services to an assigned region (area) on the earth. The power and bandwidth of these satellites depend upon the preferred size of the footprint, complexity of the traffic control protocol schemes and the cost of ground stations.
- A satellite works most efficiently when the transmissions are focused with a desired area. When the area is focused, then the emissions don't go outside that designated area and thus minimizing the interference to the other systems. This leads more efficient spectrum usage.
- Satellite's antenna patterns play an important role and must be designed to best cover the designated geographical area (which is generally irregular in shape). Satellites should be designed by keeping in mind its usability for short and long term effects throughout its life time.
- The earth station should be in a position to control the satellite if it drifts from its orbit it is subjected to any kind of drag from the external forces.

SATELLITES ORBITS

- Depending on the application, satellites orbit around the earth can be circular or elliptical. Satellites in circular orbits always keep the same distance to the earth's surface following a simple law:

The attractive force F_g of the earth due to gravity is given mathematically as:

$$F_g = m \times g \ \times (R/r)\ 2$$

The centrifugal force F_c trying to pull the satellite away equals

$$m \times r \times \omega^2$$

Where:

m is the mass of the satellite

R is the radius of earth with $R \ = \ 6{,}370\ km$

r is the distance of the satellite to the centre of the earth

g is the acceleration of gravity with g = 9.81 m/s2

ω is the angular velocity with $\omega = 2\pi f$, f is the frequency of the rotation.

To keep the satellite in a stable circular orbit, the following equation must hold:

$Fg = Fc$, i.e., both forces must be equal. Looking at this equation the first thing to notice is that the mass m of a satellite is irrelevant (it appears on both sides of the equation). Solving the equation for the distance r of the satellite to the centre of the earth results in the following equation:

The distance $r = [g \times R^2/(2\pi f)^2]^{1/3}$

- From the above equation it can be concluded that the distance of a satellite to the earth's surface depends on its rotation frequency.
- Important parameters in satellite communication are the inclination and elevation angles. The inclination angle δ (fig. 30 (a)) is defined between the equatorial plane and the plane described by the satellite orbit. An inclination angle of 0 degrees means that the satellite is exactly above the equator. If the satellite does not have a circular orbit, the closest point to the earth is called the perigee.
- The elevation angle ε (fig. 30 (b)) is defined between the centre of the satellite beam and the plane tangential to the earth's surface. A so called footprint can be defined as the area on earth where the signals of the satellite can be received.

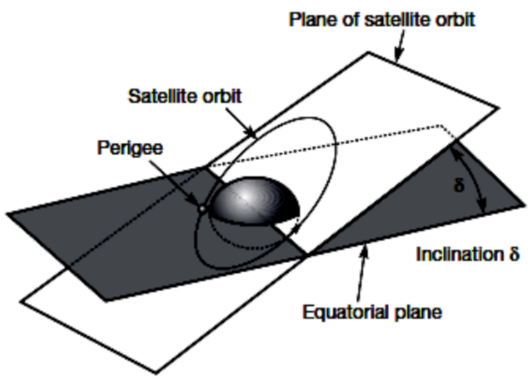


Figure 30 (a): Angle of Inclination

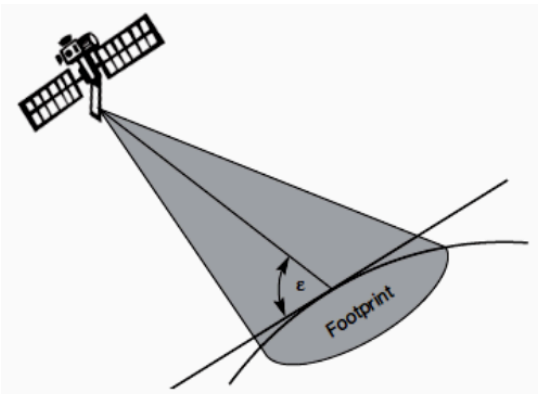


Figure 30 (b) Angle of Elevation

TYPES OF SATELLITES (BASED ON ORBITS)

GEOSTATIONARY OR GEOSYNCHRONOUS EARTH ORBIT (GEO)

GEO satellites are synchronous with respect to earth. Looking from a fixed point from Earth, these satellites appear to be stationary. These satellites are placed in the space in such a way that only three satellites are sufficient to provide connection throughout the surface of the Earth (that is; their footprint is covering almost 1/3rd of the Earth). The orbit of these satellites is circular.

There are three conditions which lead to geostationary satellites. Lifetime expectancy of these satellites is 15 years.

i. The satellite should be placed 37,786 kms (approximated to 36,000 kms) above the surface of the earth.
ii. These satellites must travel in the rotational speed of earth, and in the direction of motion of earth, that is eastward.
iii. The inclination of satellite with respect to earth must be 00.

- Geostationary satellite in practical is termed as geosynchronous as there are multiple factors which make these satellites shift from the ideal geostationary condition.

1. Gravitational pull of sun and moon makes these satellites deviate from their orbit. Over the period of time, they go through a drag. (Earth‟s gravitational force has no effect on these satellites due to their distance from the surface of the Earth.)
2. These satellites experience the centrifugal force due to the rotation of Earth, making them deviate from their orbit.
3. The non-circular shape of the earth leads to continuous adjustment of speed of satellite from the earth station.

- These satellites are used for TV and radio broadcast, weather forecast and also, these satellites are operating as backbones for the telephone networks.

Disadvantages of GEO:

Northern or southern regions of the Earth (poles) have more problems receiving these satellites due to the low elevation above a latitude of 60°, i.e., larger antennas are needed in this case. Shading of the signals is seen in cities due to high buildings and the low elevation further away from the equator limit transmission quality. The transmit power needed is relatively high which causes problems for battery powered devices. These satellites cannot be used for small mobile phones. The biggest problem for voice and also data communication is the high latency as without having any handovers, the signal has to at least travel 72,000 kms. Due to the large footprint, either frequencies cannot be reused or the GEO satellite needs special antennas focusing on a smaller footprint. Transferring a GEO into orbit is very expensive.

LOW EARTH ORBIT (LEO) SATELLITES:

- These satellites are placed 500-1500 kms above the surface of the earth. As LEOs circulate on a lower orbit, hence they exhibit a much shorter period that is 95 to 120 minutes. LEO systems try to ensure a high elevation for every spot-on earth to provide a high quality communication link. Each LEO satellite will only be visible from the earth for around ten minutes.
- Using advanced compression schemes, transmission rates of about 2,400 bit/s can be enough for voice communication. LEOs even provide this bandwidth for mobile terminals with Omni- directional antennas using low transmit power in the range of 1W. The delay for packets delivered via a LEO is relatively low (approx 10 ms). The delay is comparable to long-distance wired connections (about 5–10 ms). Smaller footprints of LEOs allow for better frequency reuse, similar to the concepts used for cellular networks. LEOs can provide a much higher elevation in Polar Regions and so better global coverage.

- These satellites are mainly used in remote sensing an providing mobile communication services (due to lower latency).

Disadvantages LEO:

- The biggest problem of the LEO concept is the need for many satellites if global coverage is to be reached. Several concepts involve 50–200 or even more satellites in orbit. The short time of visibility with a high elevation requires additional mechanisms for connection handover between different satellites.
- The high number of satellites combined with the fast movements resulting in a high complexity of the whole satellite system. One general problem of LEOs is the short lifetime of about five to eight years due to atmospheric drag and radiation from the inner Van Allen belt1. Assuming 48 satellites and a lifetime of eight years, a new satellite would be needed every two months.
- The low latency via a single LEO is only half of the story. Other factors are the need for routing of data packets from satellite to if a user wants to communicate around the world. Due to the large footprint, a GEO typically does not need this type of routing, as senders and receivers are most likely in the same footprint.

MEDIUM EARTH ORBIT (MEO) SATELLITES:

- MEOs can be positioned somewhere between LEOs and GEOs, both in terms of their orbit and due to their advantages and disadvantages. Using orbits around 10,000 km, the system only requires a dozen satellites which is more than a GEO system, but much less than a LEO system. These satellites move more slowly relative to the earth‟s rotation allowing a simpler system design (satellite periods are about six hours). Depending on the inclination, a MEO can cover larger populations, so requiring fewer handovers.

  Disadvantages MEO:

  - Again, due to the larger distance to the earth, delay increases to about 70–80 ms. the satellites need higher transmit power and special antennas for smaller footprints.

The above three are the major three categories of satellites, apart from these, the satellites are also classified based on the following types of orbits: Other example of satellites based on orbits include: **Sun- Synchronous Orbits satellites, Hohmann Transfer Orbit, Prograde orbit, Retrograde orbit, and Polar Orbits.**

<u>Typical Examples of Satellites System</u>

INTELSAT

International Telecommunication Satellite: Created in 1964

- Over 140 member countries
- More than 40 investing entities
- Early Bird satellite in 1965
- Six (6) evolutions of INTELSAT satellites between 1965-87 Geostationary orbit
- Covers 3 regions:

Atlantic Ocean Region (AOR), Indian Ocean Region (IOR), and Pacific Ocean Region (POR)

<u>U.S DOMSATS</u>

Domestic Satellite:

- In geostationary orbit
- Over 140 member countries
- Direct-to-home TV service
- Three (3) categories of U. S. DBS system: high power, medium, and low power.
- Measure in equivalent isotropic radiated power (EIRP).
- The upper limit of EIRP:

High power (60 dBW), Medium (48 dBW), and Low power (37 dBW).

POLAR ORBITING SATELLITES

These satellites follow the Polar Orbits. An infinite number of polar orbits cover north and south polar regions.

- Weather (ultraviolet sensor also measure ozone level) satellites between 800 and 900 km

- National Oceanic and Atmospheric Administration (NOAA) operate a weather satellite system
- Satellite period is 102 minutes and earth rotated 25 ^{O}.
- Estimate the sub-satellite point at the following times after the equator 90 ^{O} E North-South crossing:
  a. 10 minutes, 87.5 ^{O} E and 36 ^{O} S.
  b. 102 minutes, 65 ^{O} E and equator.
  c. 120 minutes, 60 ^{O} E and 72 ^{O} S.

- The system uses both geostationary operational environment satellite (GOES) and polar operational environment satellite (POES)
  - Sun synchronous: they across the equator at the same local time each day
  - The morning orbit, at an altitude of 830 km, crosses the equator from south to north at 7:30 AM, and the afternoon orbit, at an altitude of 870 km, at 1:40 PM.
- Search and rescue (SAR) satellite: Cospas-Sarsat.

APPLICATIONS OF SATELLITES SYSTEM

Satellites are designed for specific purposes, the following include the general applications of satellites communication system, these includes:

1. Weather Forecasting
2. Radio and TV Broadcast
3. Military Satellites
4. Navigation Satellites
5. Global Telephone

Weather Forecasting

Certain satellites are specifically designed to monitor the climatic conditions of earth. They continuously monitor the assigned areas of earth and predict the weather conditions of that region. This is done by taking images of earth from the satellite. These images are transferred using assigned radio frequency to the earth station. (Earth Station: it's a radio station located on the earth and used for relaying signals from satellites.) These satellites are exceptionally useful in predicting disasters like hurricanes, and monitor the changes in the Earth's vegetation, sea state, ocean colour, and ice fields.

Radio and TV Broadcast

These dedicated satellites are responsible for making 100s of channels across the globe available for everyone. They are also responsible for broadcasting live matches, news, world-wide radio services. These satellites require a 30-40 cm sized dish to make these channels available globally.

Military Satellites

These satellites are often used for gathering intelligence, as a communications satellite used for military purposes, or as a military weapon. A satellite by itself is neither military nor civil. It is the kind of payload it carries that enables one to arrive at a decision regarding its military or civilian character.

Navigation Satellites

The system allows for precise localization world-wide, and with some additional techniques, the precision is in the range of some meters. Ships and aircraft rely on GPS as an addition to traditional navigation systems. Many vehicles come with installed GPS receivers. This system is also used, e.g., for fleet management of trucks or for vehicle localization in case of theft.

Global Telephone

One of the first applications of satellites for communication was the establishment of international telephone backbones. Instead of using cables it was sometimes faster to launch a new satellite. But, fiber optic cables are still replacing satellite communication across long distance as in fiber optic cable, light is used instead of radio frequency, hence making the communication much faster (and of course, reducing the delay caused due to the amount of distance a signal needs to travel before reaching the destination.).

Using satellites, to typically reach a distance approximately 10,000 kms away, the signal needs to travel almost 72,000 kms, that is, sending data from ground to satellite and (mostly) from satellite to another location on earth. This cause‟s substantial amount of delay and this delay becomes more prominent for users during voice calls.

Connecting Remote Areas

Due to their geographical location many places all over the world do not have direct wired connection to the telephone network or the internet (e.g., researchers on Antarctica) or because of the current state of the infrastructure of a country. Here the satellite provides a complete coverage and (generally) there is one satellite always present across a horizon.

Global Mobile Communication

The basic purpose of satellites for mobile communication is to extend the area of coverage. Cellular phone systems, such as AMPS and GSM (and their successors) do not cover all parts of a country. Areas that are not covered usually have low population where it is too expensive to install a base station. With the integration of satellite communication, however, the mobile phone can switch to satellites offering world-wide connectivity to a customer. Satellites cover a certain area on the earth. This area is termed as a „footprint‟ of that satellite. Within the footprint, communication with that satellite is possible for mobile users. These users communicate using a Mobile-User-Link (MUL). The base-stations communicate with satellites using a Gateway-Link (GWL). Sometimes it becomes necessary for satellite to create a communication link between users belonging to two different footprints. Here the satellites send signals to each other, and this is done using Inter-Satellite-Link (ISL).

FREQUENCY ALLOCATION FOR SATELLITE

Recall the frequency allocation for RADAR system as discussed in the previous topic, satellites system also has a designated frequency allocation.

- ITU (International Telecommunication Union) allocated a portion of these bands for Satellites communication system as well.

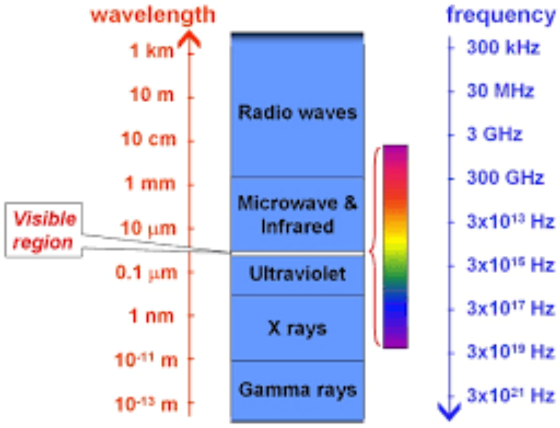


Fig. 31 Electromagnetic spectrum Wavelength and corresponding frequencies.

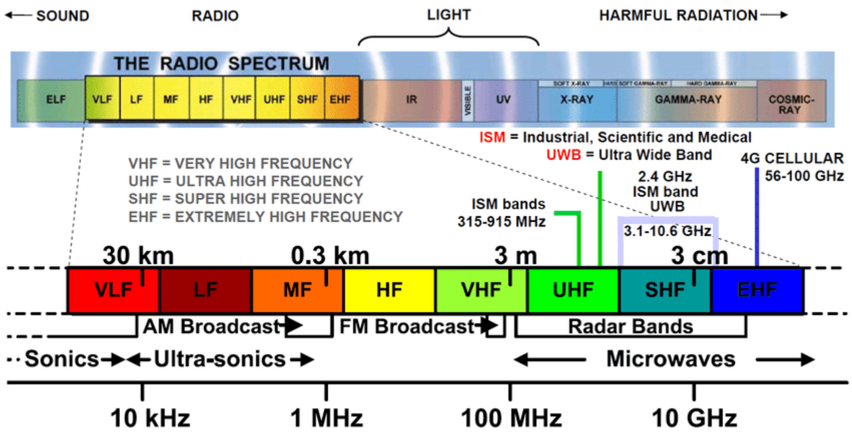


Fig. 32 Electromagnetic Radio spectrum

Allocation of frequencies to satellite services is a complicated process which requires international coordination and planning. This is done as per the ITU. To implement this frequency planning, the world is divided into three regions:

1. Region1: Europe, Africa, and Mongolia
2. Region 2: North and South America and Greenland
3. Region 3: Asia (excluding region 1 areas), Australia and south-west Pacific.

- Within these regions, the HE frequency bands are allocated to various satellite services. Some of them are listed below.
  - Fixed satellite service: Provides Links for existing Telephone Networks Used for transmitting television signals to cable companies
  - Broadcasting satellite service: Provides Direct Broadcast to homes. E.g. Live Cricket matches etc
  - Mobile satellite services: This includes services for: Land Mobile

  - Maritime Mobile Aeronautical mobile
  - Navigational satellite services: Include Global Positioning systems
  - Meteorological satellite services: They are often used to perform Search and Rescue service

- Below are the frequencies allocated to these satellites: Frequency Band (GHZ) Designations:
    - VHF: 01-0.3
    - UHF: 0.3-1.0
    - L-band: 1.0-2.0
    - S-band: 2.0-4.0
    - C-band: 4.0-8.0
    - X-band: 8.0-12.0
    - Ku-band: 12.0-18.0 (Ku is Under K Band)
    - Ka-band: 18.0-27.0 (Ka is Above K Band)
    - V-band: 40.0-75.0
    - W-band: 75-110
    - mm-band: 110-300
    - µm-band: 300-3000
- Based on the satellite service, following are the frequencies allocated to the satellites:

Frequency Band (GHZ)	**Designations:**
VHF: 01-0.3	Mobile & Navigational Satellite Services
L-band: 1.0-2.0	Mobile & Navigational Satellite Services
C-band: 4.0-8.0	Fixed Satellite Service
Ku-band: 12.0-18.0	Direct Broadcast Satellite Services

SATELLITES PROBLEMS AND SOLUTIONS

Reading Assignment V